PHYS 7501:
Particles and Nuclei I

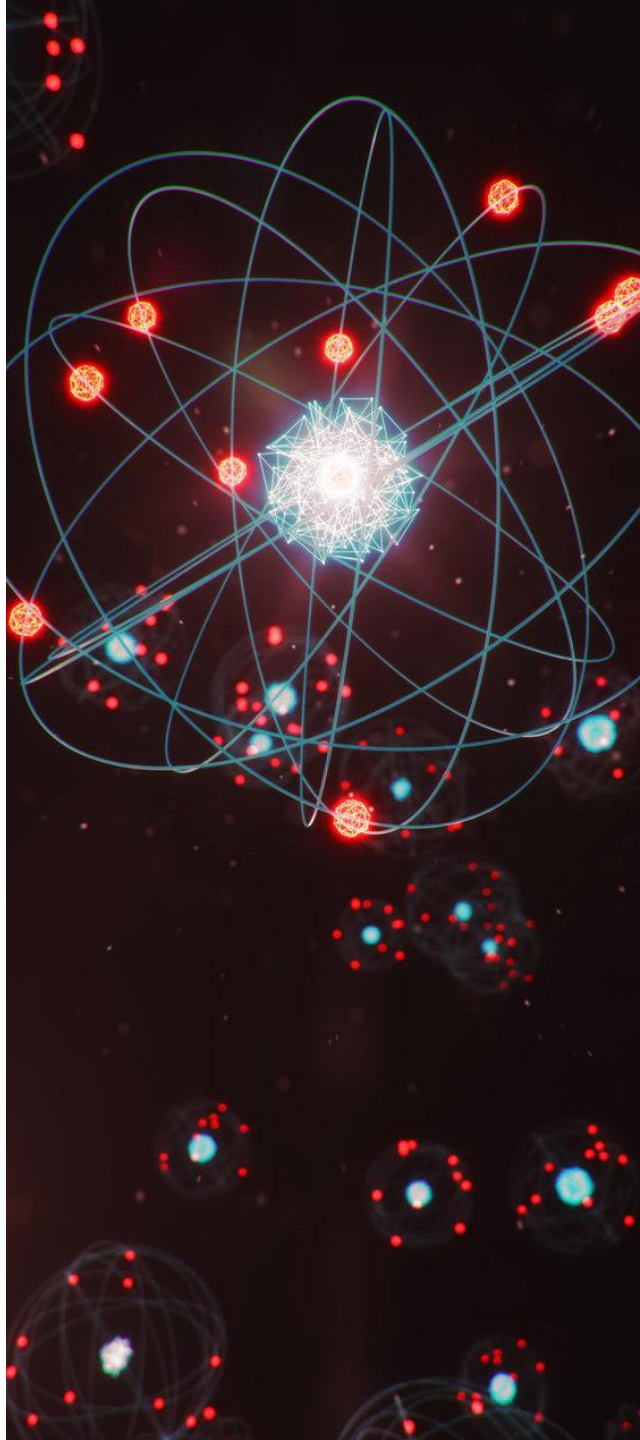
Tues./Thurs. 2 – 3:20PM
Clippinger 105

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Available via e-mail & Teams



MID-TERM REVIEW

This won't cover necessarily everything but a broad overview!

Monday	Tuesday	Wednesday	Thursday	Friday
9/29	9/30	10/1	10/2	10/3
	Gamma Decay		**Review ☺**	
10/6	10/7	10/8	10/9	10/10
	Exam Day		**Fission**	*Fall Break*
10/13	10/14	10/15	10/16	10/17
	Nuclear Reactions HW#3 Due		**Scattering**	

How big is a nucleus?

Phenomenological estimates:

- A nucleus' mass is roughly: M(Z,A) = A x amu
  - **1 amu = atomic mass unit = u = 931.494 MeV/c^2 ≅ 1.66 x 10^{-24} g**
  - The amu is defined so that M(^{12}C) = 12u

  - A nucleus' radius is roughly: R(Z,A) = (1.2fm) x $A^{1/3}$
    - fm = femtometer (aka fermi) = 10^{-15} m
    - The radius of a nucleon is often referred to as r_0 = 1.2 fm

  - Therefore, an estimate of the nuclear density is:
    - $n = \frac{\#}{V} = \frac{\#}{\frac{4}{3}\pi R^3} = \frac{\#}{\frac{4}{3}\pi(1.2fm)^3 A} \approx 0.14\ nucleons/fm^3$
    - This seems small but when you convert to mass-density, this is 2x10^{14} g/cm^3

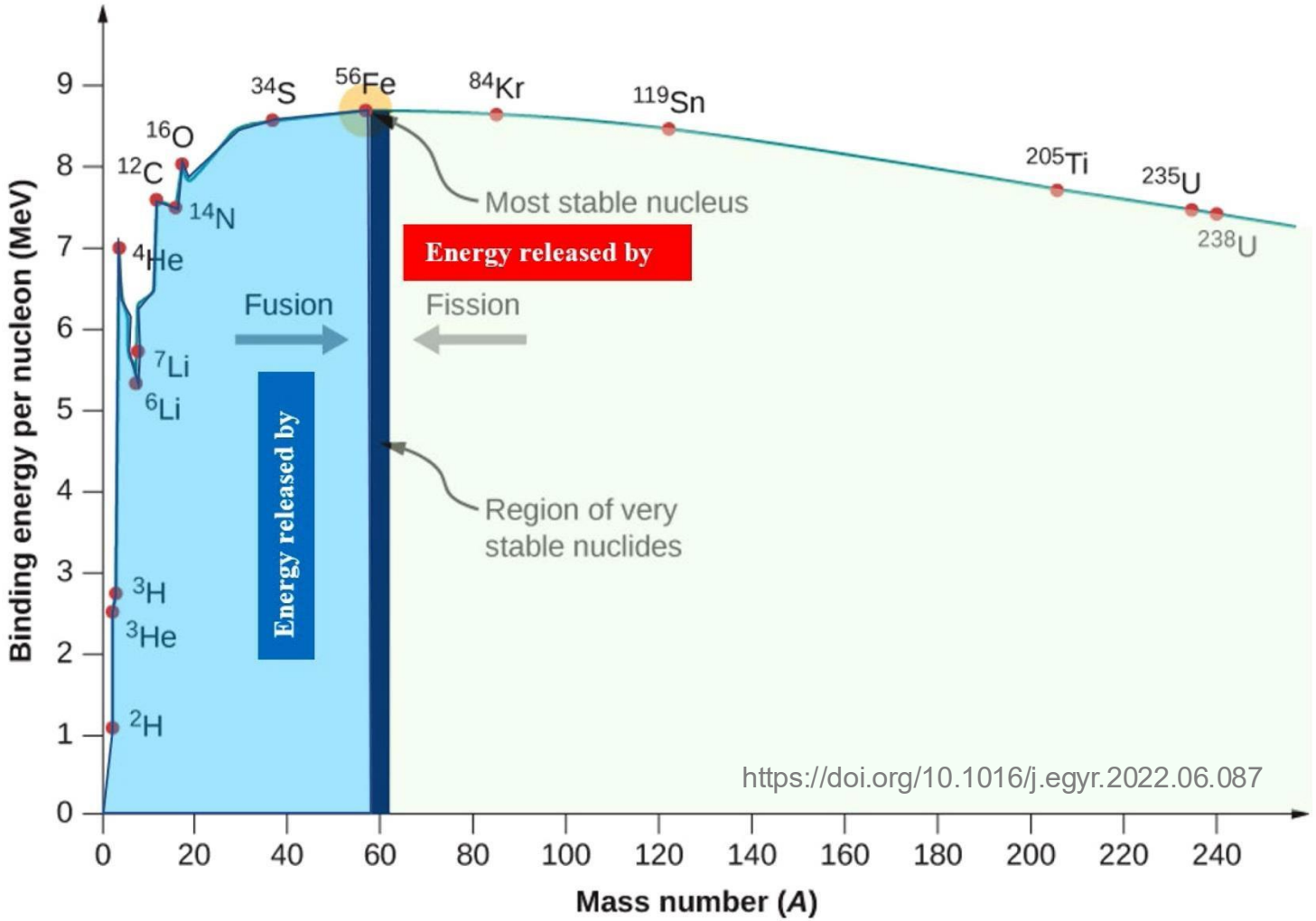
Nuclear Transmutation

- Rules for converting one nuclide (or nuclides) to another (or others)
  - Charge Conservation: $\sum q_{before} = \sum q_{after}$ (q from protons + positrons + electrons)
  - Baryon Conservation: $\sum A_{before} = \sum A_{after}$ (A from neutrons + protons)
  - Lepton Number Conservation: $[N_{leptons} - N_{anti-leptons}]_{before} = [N_{leptons} - N_{anti-leptons}]_{after}$
- Likelihood of transmutation is impacted by energetics and spin/parity selection rules

- <u>Two Types</u>:
  - **Reactions**:
    - Multiple reactants create one or more products
    - Notation: A + b → c + D is written as A(b,c)D, where M(b) < M(A) and M(c) < M(D)
      - Example: $^{12}C + \alpha \rightarrow {}^{16}O + \gamma$ is $^{12}C(\alpha,\gamma)^{16}O$ and is called "carbon twelve alpha gamma"
  - **Decays**:
    - α, β^+, β^-, e^--capture, β-delayed γ/p/α/n emission, fission, cluster emission, prompt γ
    - All lose nucleons except β-decays, e^- capture, and prompt γ (following a reaction)

Binding Energy

- Average BE/A is 8 MeV per nucleon (MeV/u or MeV/A)
- ^{56}Fe is the most bound nucleus
  - This is why nuclear fusion in star cores ends with iron!
- At low A, attractive forces between nucleons dominate over repulsive electrostatic forces.
- At high A, repulsive electrostatic forces begin to dominate, and these forces tend to break apart the nucleus rather than hold it together.



Nuclear Masses and Energy

- Nuclear Mass: $\boldsymbol{m(Z,N) = Zm_p + Nm_n - BE}$
- Atomic Mass: $\mathrm{M}(\boldsymbol{Z,N}) = \boldsymbol{Z(m_p + m_e) + Nm_n - BE}$

- It's often easier to use mass excess, denoted as ME or Δ
  - The difference between atomic mass and $\mathrm{A} \times \mathrm{u}$: $\Delta M(A,Z) = M(Z,N) - A \times u$
  - Defined so that ^{12}C has a mass excess of zero
  - Look up values on NuBase (NNDC)

- Example: Mass excess of ^{64}Cu:
  - $\Delta M(^{64}\mathrm{Cu}) = 63.929764\mathrm{u} - 64 \times u = -65.42\ \mathrm{MeV}$ (recall that $1u$ = 931.5 MeV/c^2)

Q-values

Recall our transmutations:
Reactions: A + b → c + D
Decay: A → B + particle

- Energy change in nuclear reactions is called the **Q-value**

$$Q = \sum_{reactants} \Delta(A,Z) - \sum_{products} \Delta(A,Z)$$

^{68}Se + p → ^{69}Br + γ
reactants *products*

- For example, ^{68}Se(p,γ)^{69}Br

$$Q = \Delta(^{68}Se) + \Delta(p) - \Delta(^{69}Br)$$

$$= -54.189\ MeV - 7.288\ MeV - -46.260\ MeV$$

$$= -0.641\ \text{MeV}$$

^{68}Se + p

Reaction Products	Q-Value (keV)	
^{69}Br + γ	-6.4E+2	*4*

Separation Energies

- **Separation Energy** of a nucleon is the minimum energy required to remove a single nucleon from a nucleus.

- One Proton Separation Energy:

$$S_p(Z,N) = \Delta(Z-1,N) + \Delta(p) - \Delta(Z,N)$$

- Two Proton Separation Energy:

$$S_{2p}(Z,N) = \Delta(Z-2,N) + 2 \times \Delta(p) - \Delta(Z,N)$$

- One Neutron Separation Energy:

$$S_n(Z,N) = \Delta(Z,N-1) + \Delta(n) - \Delta(Z,N)$$

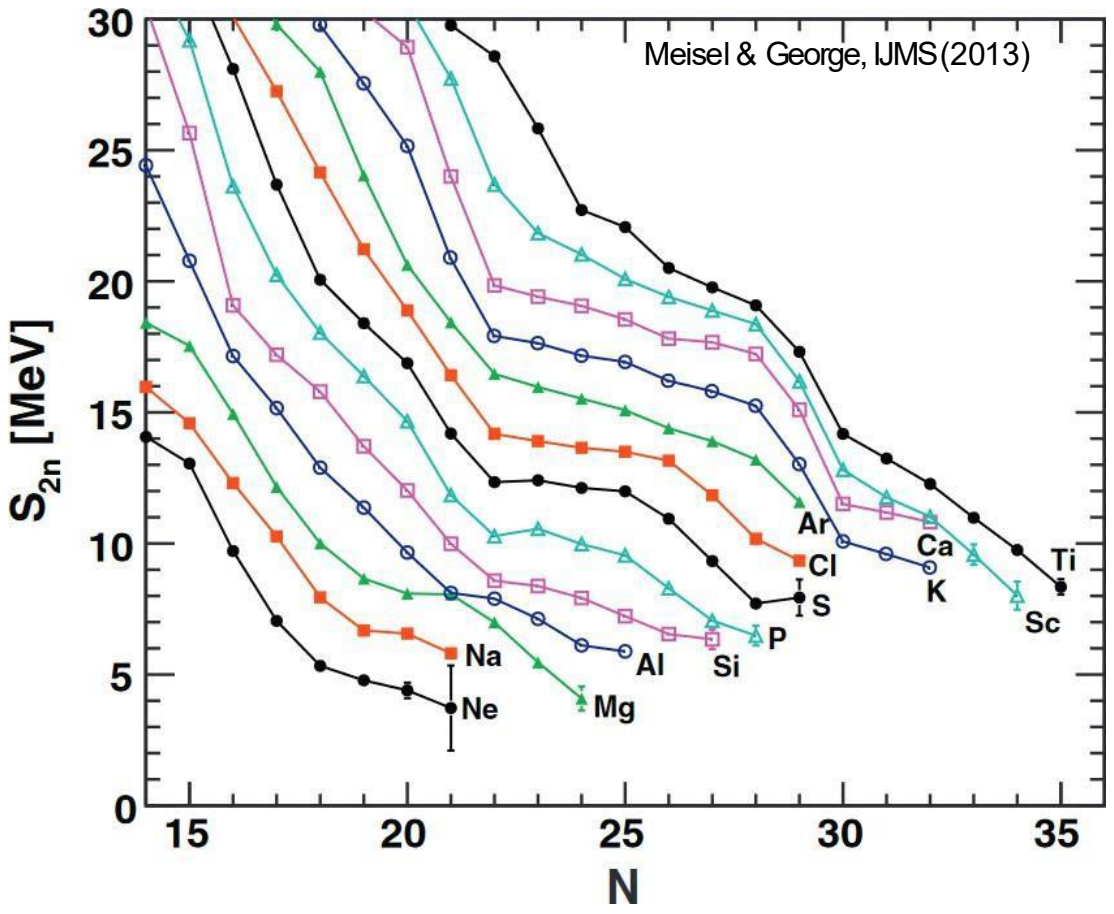
- Two Neutron Separation Energy:

$$S_{2n}(Z,N) = \Delta(Z,N-2) + 2 \times \Delta(n) - \Delta(Z,N)$$

Separation Energy and Driplines

- A positive separation energy means the nucleus is stable against particle emission
- A negative separation energy means the nucleus is unstable against particle emission

- Proton and neutron driplines are the boundaries of the Nuclear Chart at the point where the separation energy becomes zero or negative.
  - Beyond this, the nucleus is unbound and can ”drip” a proton or neutron



Semi-Empirical Mass Formula

- C. von Weizsäcker developed the first theory for nuclear masses (Z. Phys. A 1935), now known as the semi-empirical mass formula (SEMF)
  - Also known as the Weizsäcker formula, Bethe–Weizsäcker formula, or Bethe–Weizsäcker mass formula
  - This is a parameterization of the total binding energy of the nucleus
- The **Liquid Drop Model**: Assume nuclei behave as if they are incompressible uniformly charged liquid drops and approximate the nucleus as a sphere with a uniform interior density

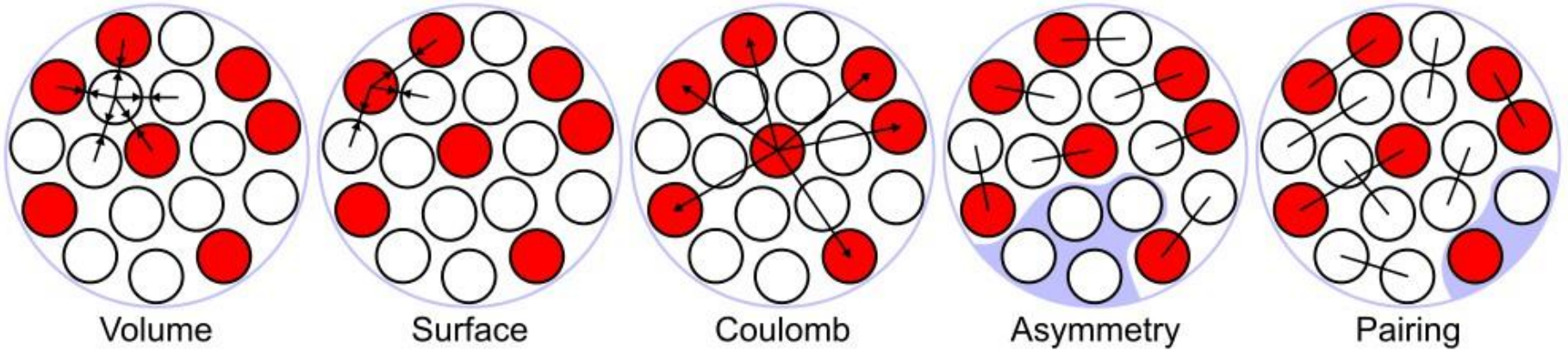


Liquid Drop:

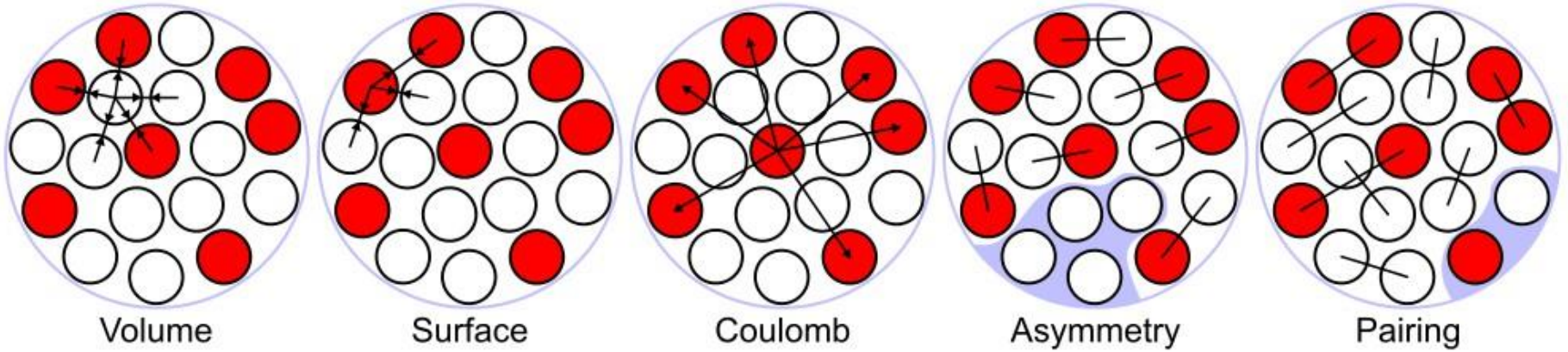
- Short-range intermolecular forces.
- Density independent of drop size.
- Heat required to evaporate fixed mass independent of drop size.

Nucleus

- Nuclear force short range.
- Density independent of nuclear size.
- B/A ~ constant



- We want to more carefully consider all interactions in a nucleus
- All nucleons are attracting each other via the Strong Force, generating some overall binding energy
- However, nucleons at the surface don't have a neighbor, reducing binding energy
- Protons repel each other due to the Coulomb force, reducing binding energy
- p-n interaction is stronger than p-p or n-n, favors N=Z but we know this only happens for a few nuclei, reducing binding energy
- Nucleons prefer to be paired, so there is an enhancement for even-even nuclei, reduction for odd-odd nuclei, and neutral for even-odd nuclei



$$BE(A,Z) = Volume - Surface - Coulomb\ - Asymmetry\ \pm Pairing$$

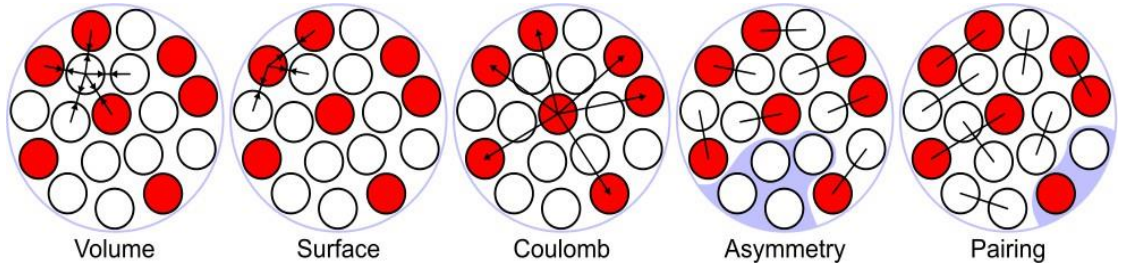
$$BE(A,Z) = a_v A - a_s A^{\frac{2}{3}} - a_c \frac{Z^2}{A^{\frac{1}{3}}} - a_a \frac{(A-2Z)^2}{A} \pm \delta$$

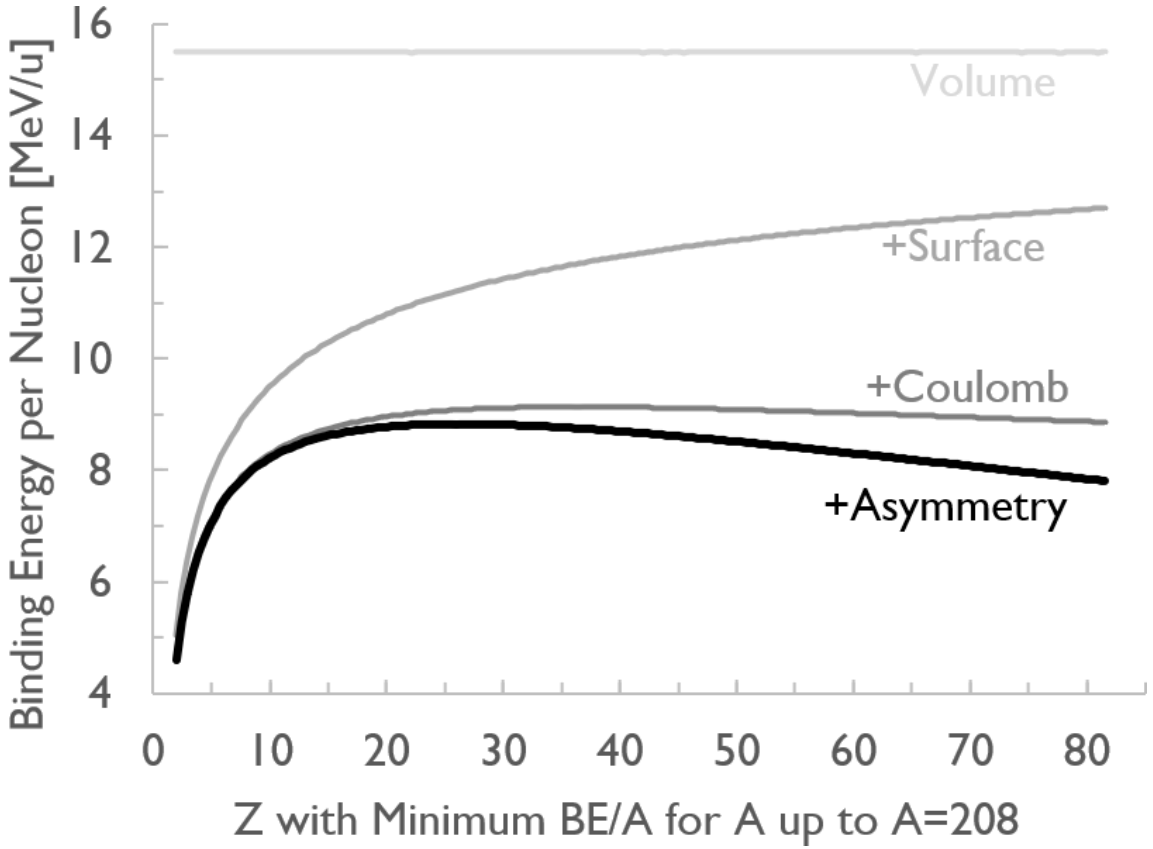
There are other forms of this equation, this is the book's notation

Volume (a_v)
Surface (a_s)
Coulomb (a_c)
Asymmetry (a_a)
Pairing (δ):
- Even-Z, Even-N: +δ
- Odd, Odd-N: -δ
- Even-Z, Odd-N: 0

a_i values are fit to empirical mass data

SEMF





Volume (a_v): $a_v = 15.56\ MeV$

Surface (a_s): $a_s = 17.23\ MeV$

Coulomb (a_c): $a_c = 0.7\ MeV$

Asymmetry (a_a): $a_a = 23.285\ MeV$

Pairing (δ): $\delta = \frac{11}{A^{1/2}}\ MeV$

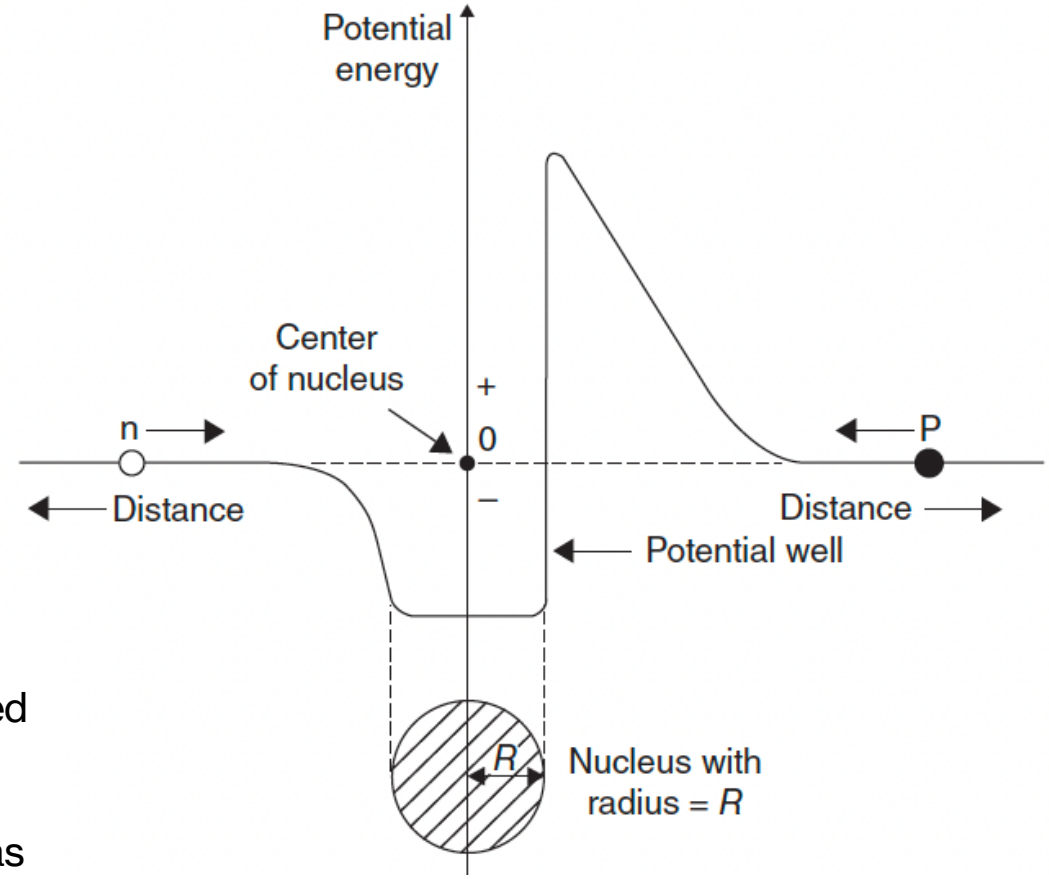
- Even-Z, Even-N: $+\delta$
- Odd, Odd-N: $-\delta$
- Even-Z, Odd-N: 0

Nuclear Potentials

- Combined interactions between neutrons and protons can be described by a ”nuclear potential well”
- Treat protons and neutrons as moving inside separate wells but superimposed on top of each other
- A simplistic view:

Protons:

- As a proton gets close to the nucleus, it is first repelled due to the long-range Coulomb force.
- As it nears the surface, the nuclear attraction (same as neutron) will overcome Coulomb repulsion. There will always be a net repulsion from other protons.
- Coulomb repulsion decreases overall attraction so the proton potential energy well will not be as deep as the neutron well.



Nuclear Potentials and States

- Understanding the potential well means that we can solve the calculation of quantum-mechanical states that "fill" the well – we usually call these "single particle levels"
- The behavior of a quantum-mechanical state is described by the wave function $\boldsymbol{\psi}$
- For some potential, we can solve for $\boldsymbol{\psi}$ using the Schrödinger equation, $\boldsymbol{H\psi = E\psi}$

$$\boldsymbol{T\psi + V\psi = E\psi} \qquad \text{or} \qquad \boldsymbol{\frac{-\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi}$$

In Cartesian coordinates,
$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

- The solutions $\boldsymbol{\psi}$ are eigenfunctions and their eigenvalues are corresponding energy $\boldsymbol{E}$
- We need to assume a potential $\boldsymbol{V}$
- For a single nucleon in the field of the nucleus,
  - $\boldsymbol{V}$ should approximate the mean-field generated by all other nucleons
  - The solutions will be single-particle levels, i.e. discrete states the nucleon can occupy
- Single nucleons are indistinguishable; we only need to solve for the single-particle levels for a nucleon and then we can fill those levels (working in terms of increasing $\boldsymbol{E}$) to generate a model to calculate the properties of our nucleus

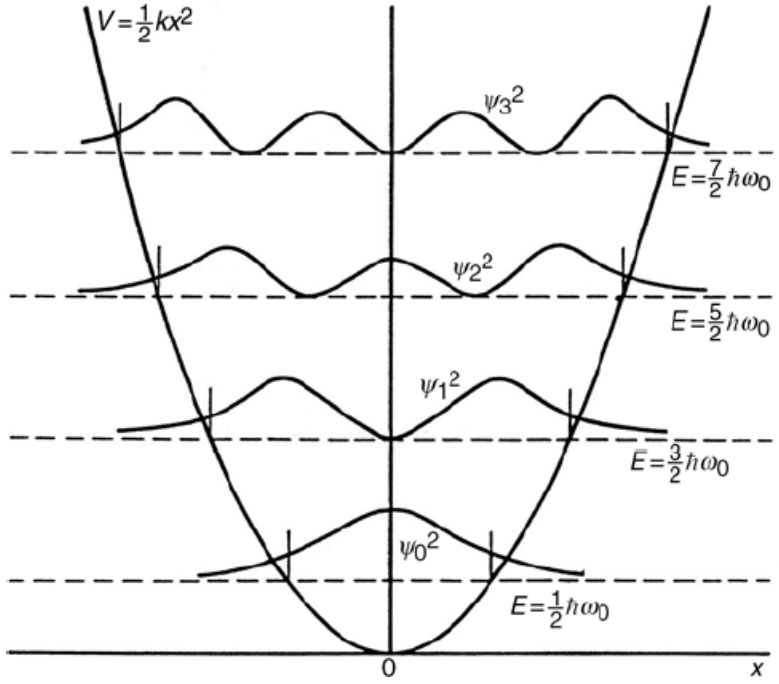
The Harmonic Oscillator Potential

- First try at the potential $\boldsymbol{V}$
- In 1935 Heisenberg actually posited that a parabolic potential could be assumed, conveniently allowing the adoption of harmonic oscillator solutions
- Since the potential doesn't saturate and has a rounded bottom, it won't be good for large nuclei with central volumes. It's mainly used for light nuclei.
- This provides evenly spaced energy levels $\boldsymbol{n \geq 1}$, with

$$\boldsymbol{E_n = (n - 1 + \frac{1}{2})\hbar\omega}$$

- The corresponding angular momenta are

$$\boldsymbol{\ell = n - 1, n - 3, \ldots \geq 0}$$

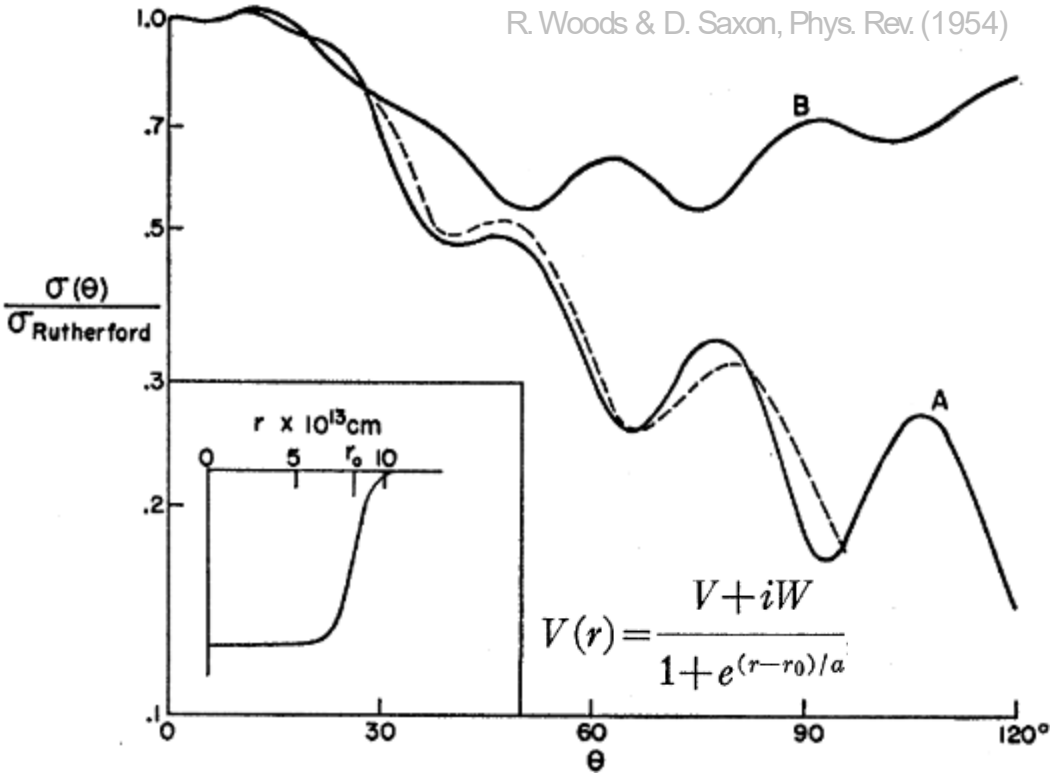


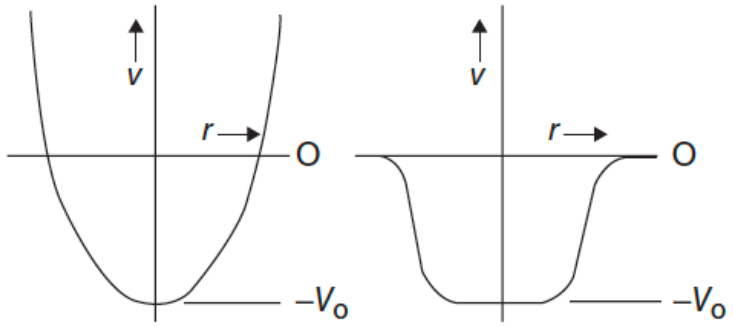
Only odd or even functions are allowed for each oscillator shell

- The number of particles per angular momentum is $\boldsymbol{2(2\ell + 1)}$ for $\boldsymbol{(2\ell + 1)}$ projections and $\boldsymbol{2}$ spins

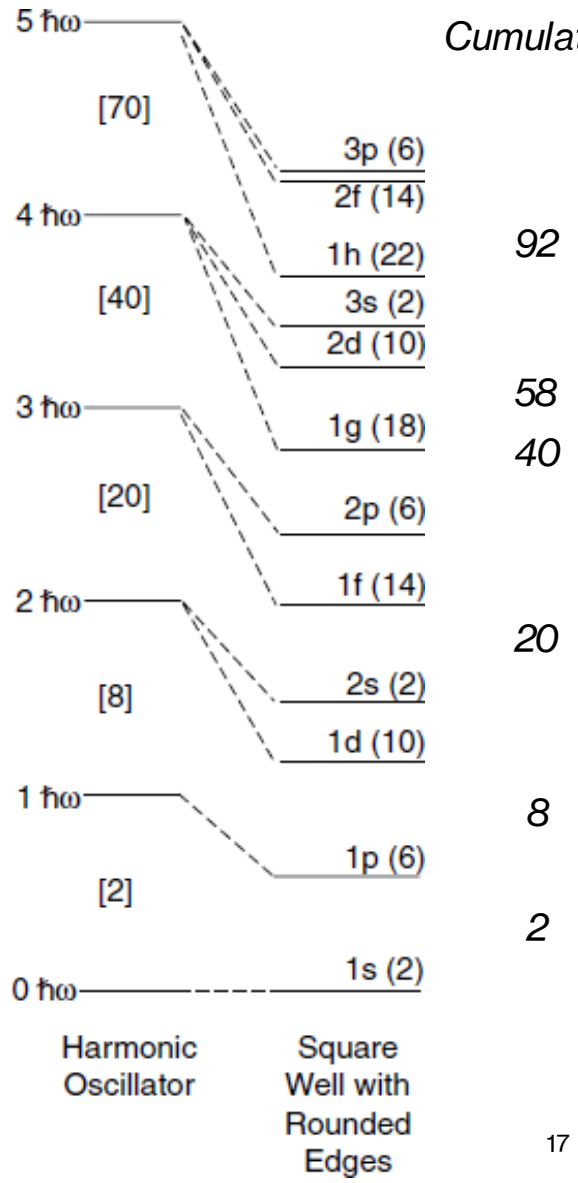
The Woods-Saxon Potential

- Since the nuclear interaction is short-range, a natural improvement would be to adopt a central potential mimicking the empirical density distribution
- This is basically a square well with soft edges, as described by the Woods-Saxon potential, an *empirical* potential!





- This is a more realistic potential, but we still don't get the Magic Numbers

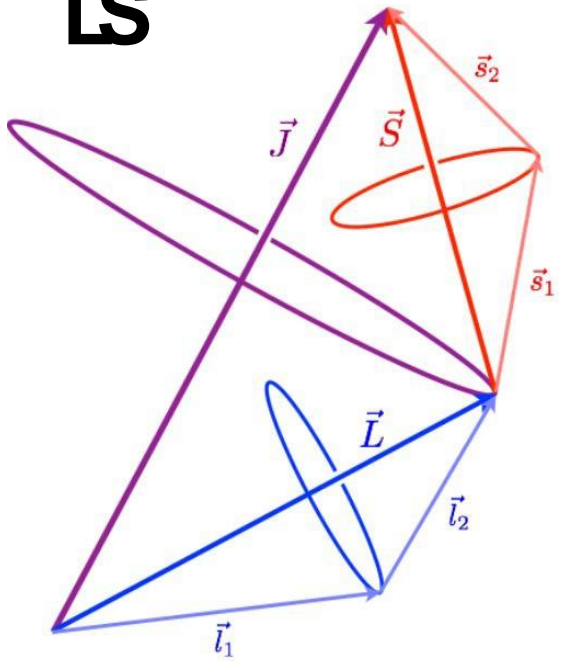


What is the missing piece?

- Relativistic particles have a tendency to align their orbital and intrinsic angular momenta
- In 1949, Maria Göppert-Mayer posited that the nucleon spin and orbital angular momentum interacted strongly making $\boldsymbol{j}$ the good quantum number for a nucleon: $\vec{\boldsymbol{j}} = \vec{\ell} + \vec{\boldsymbol{s}}$
- Prior to this, angular momentum was coupled similar to atoms where $\vec{\boldsymbol{J}} = \vec{\boldsymbol{L}} + \vec{\boldsymbol{S}}$, $\vec{\boldsymbol{L}} = \sum_{\boldsymbol{nucleons}} \vec{\ell}$, $\vec{\boldsymbol{S}} = \sum_{\boldsymbol{nucleons}} \vec{\boldsymbol{s}}$.
  - This is called *LS coupling*.
- Positing that the spin-orbit interaction is stronger than spin-spin or orbit-orbit means that instead $\vec{\boldsymbol{J}} = \sum_{\boldsymbol{nucleons}} \vec{\boldsymbol{j}}$ and $\vec{\boldsymbol{j}} = \vec{\ell} + \vec{\boldsymbol{s}}$.
  - This is called *jj coupling*.

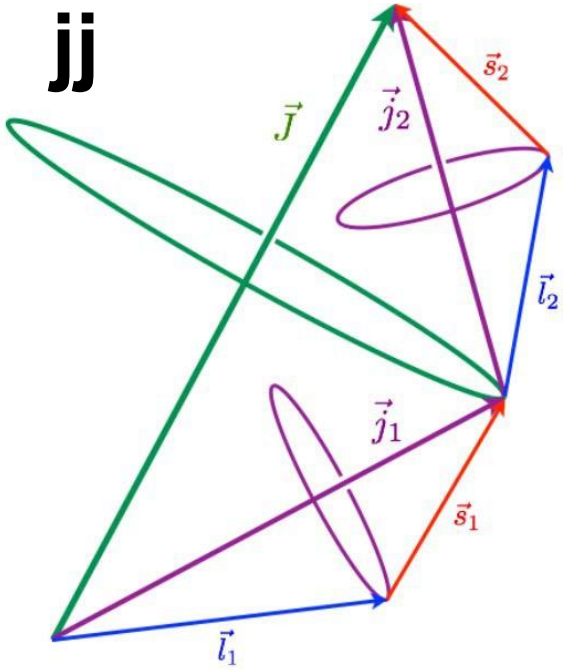
The missing piece was the spin-orbit interaction!

LS



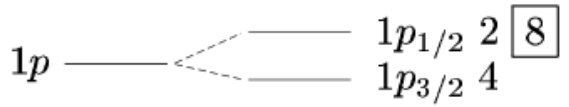
A. Kastberg, Lecture Notes Physique Atomique

jj



- So how do we calculate the $\boldsymbol{j}$ of a valence nucleon?

- $\boldsymbol{j}$ can only have values of $|\ell - \boldsymbol{s}| \leq \boldsymbol{j} \leq |\ell + \boldsymbol{s}|$ so for nucleons, $\left|\ell - \frac{\mathbf{1}}{\mathbf{2}}\right| \leq \boldsymbol{j} \leq \left|\ell + \frac{\mathbf{1}}{\mathbf{2}}\right|$ meaning the ℓ and s are either aligned $(\ell + \boldsymbol{s})$ or anti-aligned $(\ell - \boldsymbol{s})$
  - For $\ell = \mathbf{0}, \boldsymbol{j} = \frac{\mathbf{1}}{\mathbf{2}}$ $\ell = \mathbf{1}, \boldsymbol{j} = \frac{\mathbf{1}}{\mathbf{2}}\ \boldsymbol{or}\ \frac{\mathbf{3}}{\mathbf{2}}$ $\ell = \mathbf{2}, \boldsymbol{j} = \frac{\mathbf{3}}{\mathbf{2}}\ \boldsymbol{or}\ \frac{\mathbf{5}}{\mathbf{2}}$ $\ell = \mathbf{3}, \boldsymbol{j} = \frac{\mathbf{5}}{\mathbf{2}}\ \boldsymbol{or}\ \frac{\mathbf{7}}{\mathbf{2}}$
- Each $\boldsymbol{j}$ state has $\mathbf{2}\boldsymbol{j} + \mathbf{1}$ projections
  - i.e. 2 states for $\boldsymbol{j} = \frac{\mathbf{1}}{\mathbf{2}}$, 4 states for $\boldsymbol{j} = \frac{\mathbf{3}}{\mathbf{2}}$, 6 states for $\boldsymbol{j} = \frac{\mathbf{5}}{\mathbf{2}}$, …
- The spin-orbit interaction means that there is a $\boldsymbol{j}$-dependent term in the nuclear potential, so levels corresponding to different $\boldsymbol{j}$ for some ℓ will be split in energy
- For nucleons, cases with aligned $\boldsymbol{j}$ and ℓ are energetically favored,
  - For example, $\ell = \mathbf{1}, \boldsymbol{j} = \frac{\mathbf{3}}{\mathbf{2}}$ will be lower in energy than $\ell = \mathbf{1}, \boldsymbol{j} = \frac{\mathbf{1}}{\mathbf{2}}$

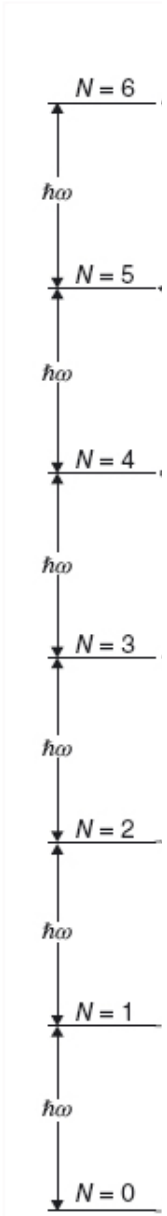


- Recall that spectroscopic notation is $\ell = \mathbf{0}, \mathbf{1}, \mathbf{2}, \mathbf{3}, \ldots = \boldsymbol{'s'}, \boldsymbol{'p'}, \boldsymbol{'d'}, \boldsymbol{'f'}, \ldots$

This work got Maria Göppert-Mayer the Nobel Prize!

a) single-particle energies of a harmonic oscillator potential as a function of the oscillator quantum number $N = 2(n-1) + \ell$ corresponding to the total number of oscillator quanta excited

- Each single-particle state N can have multiple ℓ



Harmonic oscillator

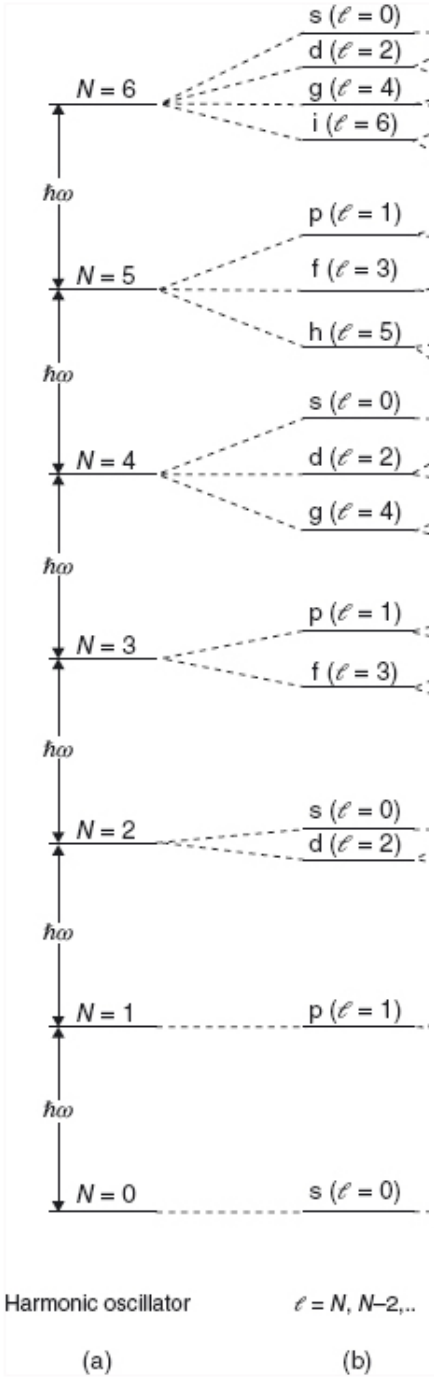
(a)

Nuclear Physics of Stars, C. Iliadis

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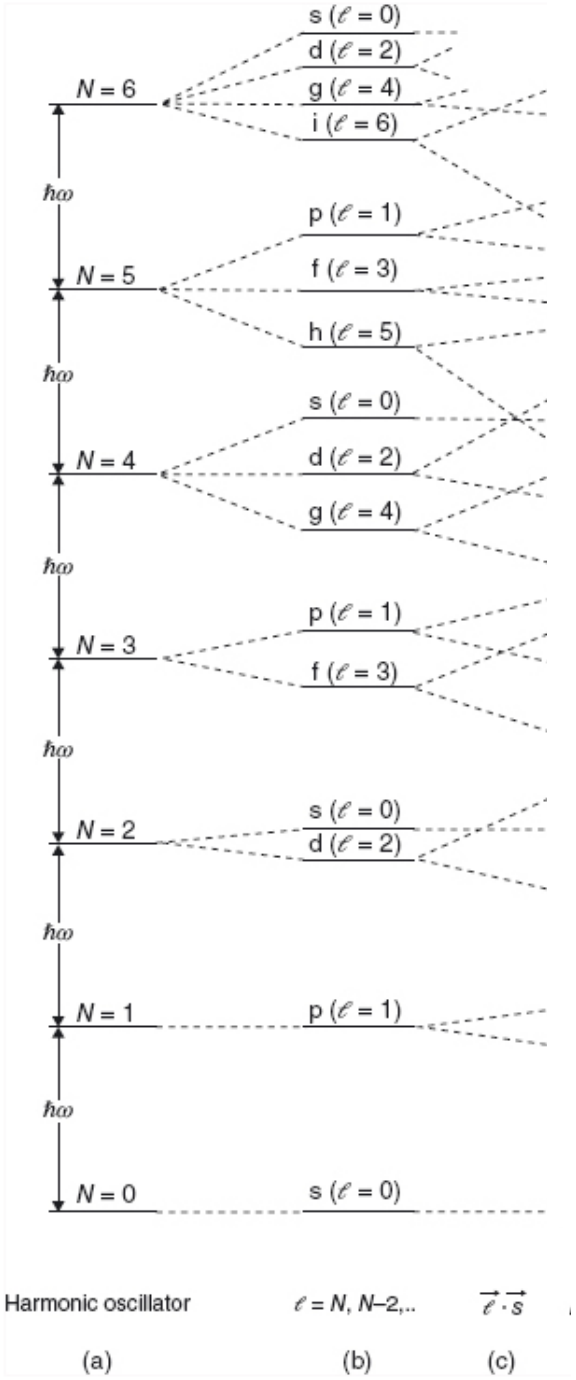
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c) Spin-orbit term splits each state with $\ell > 0$ into two states given by $j = \ell - \frac{1}{2}$ or $j = \ell + \frac{1}{2}$



Nuclear Physics of Stars, C. Iliadis

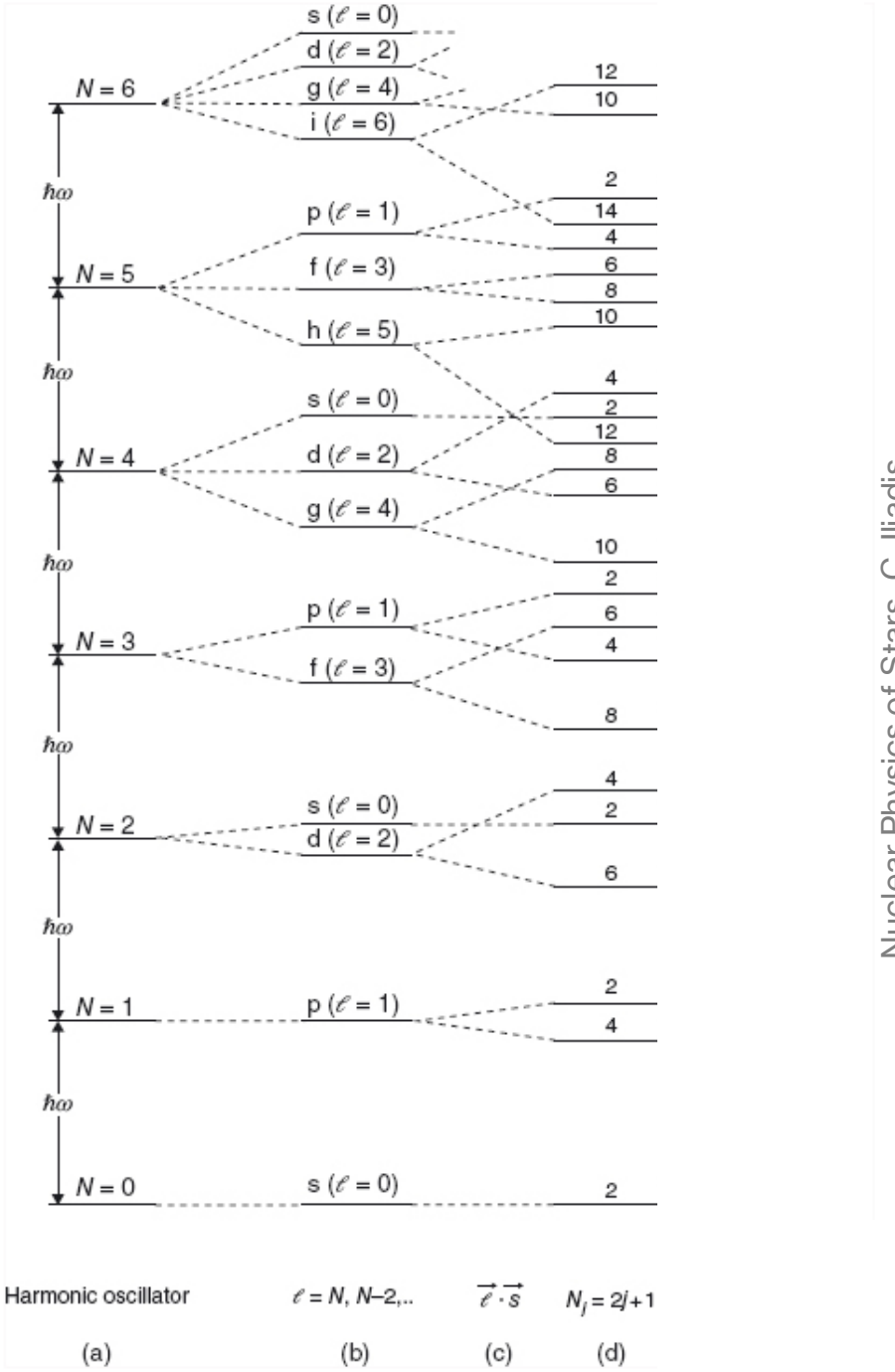
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d) Number of identical particles that can occupy a state given by $2j + 1$



Nuclear Physics of Stars, C. Iliadis

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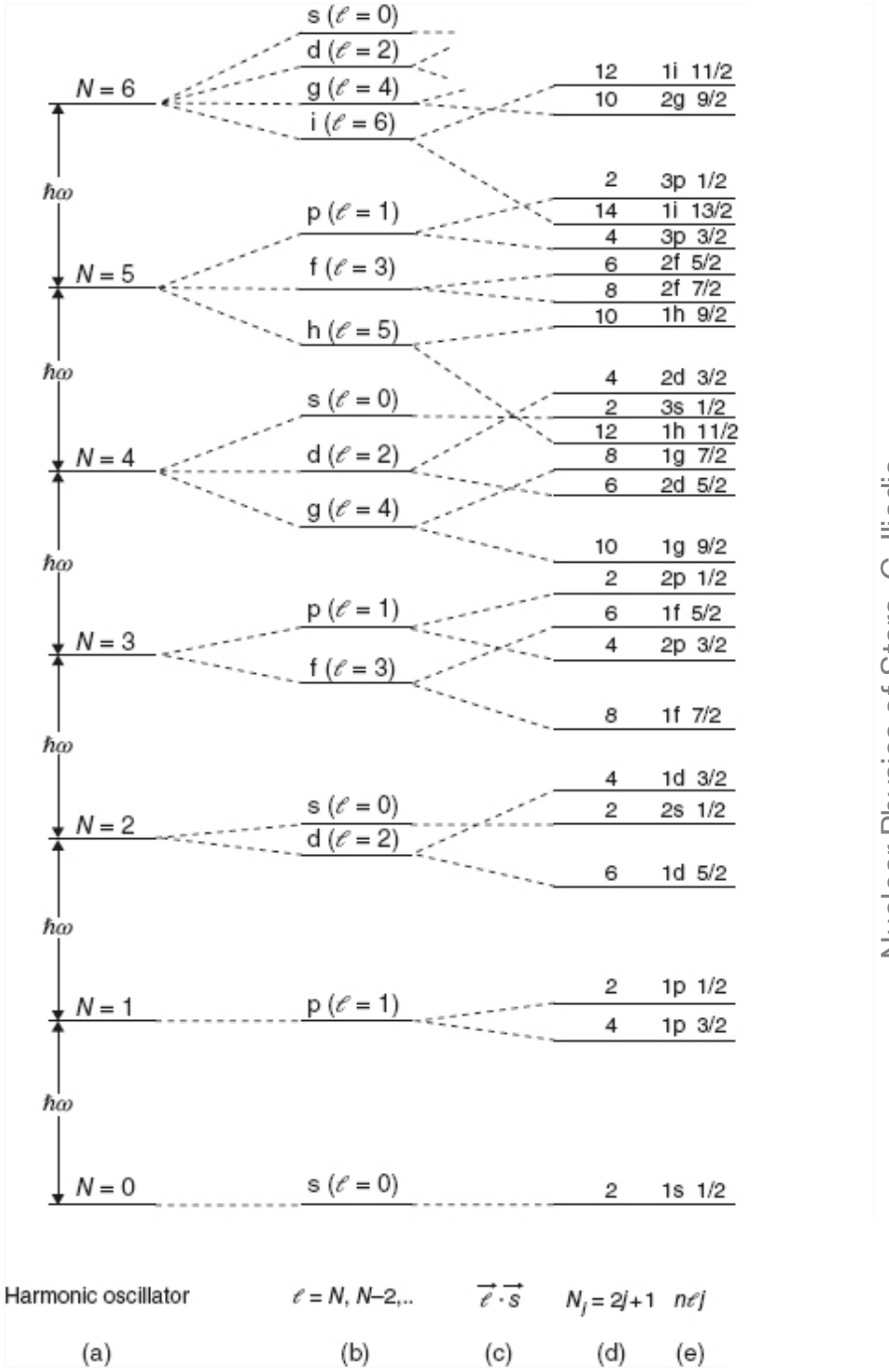
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e) Single-particle spectroscopic notation $n\ell j$



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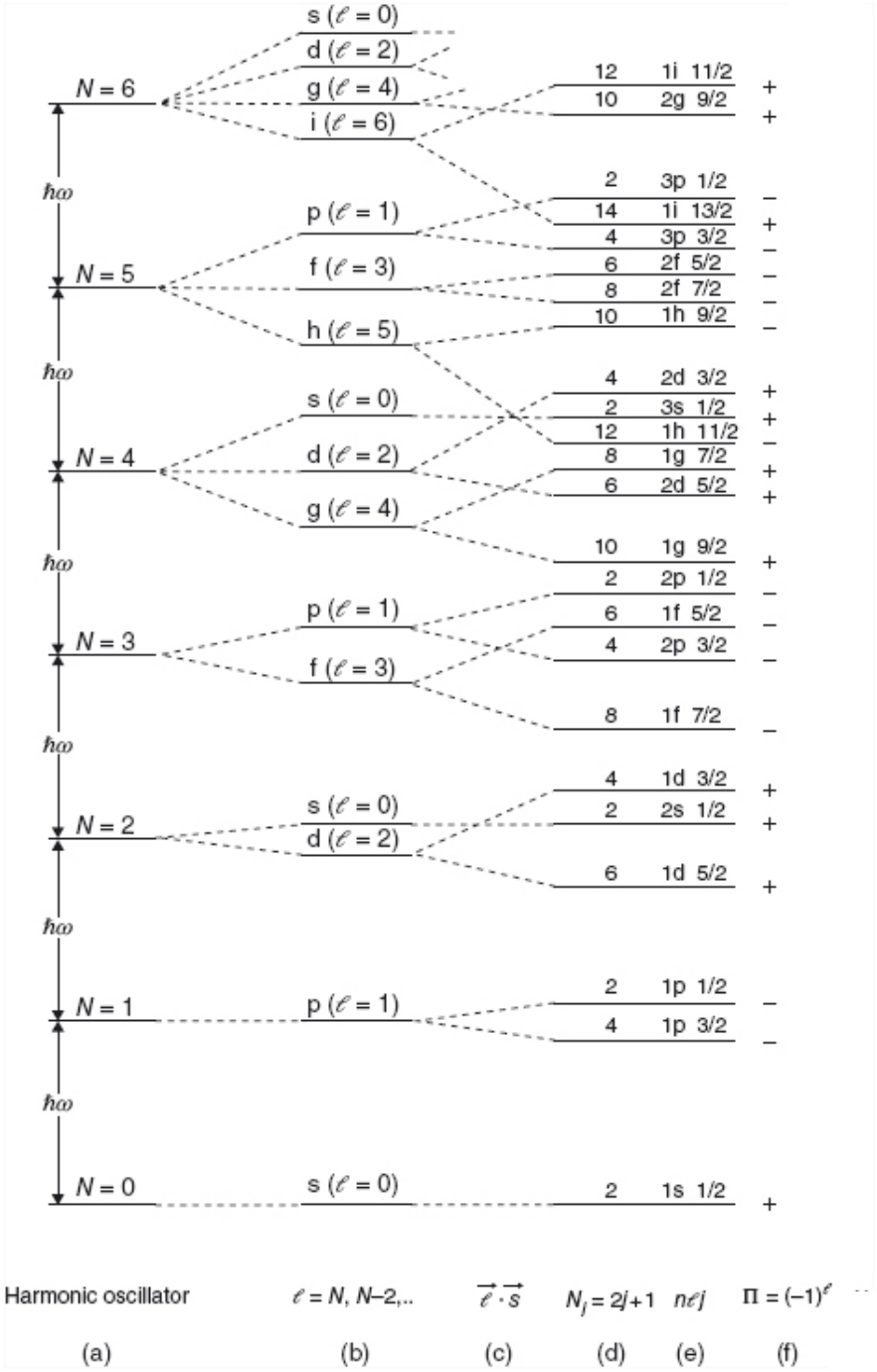
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f) Parities of single-particle levels given by $\pi = (-1)^{\ell}$



Nuclear Physics of Stars, C. Iliadis

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b) single-particle energies of a Woods-Saxon potential

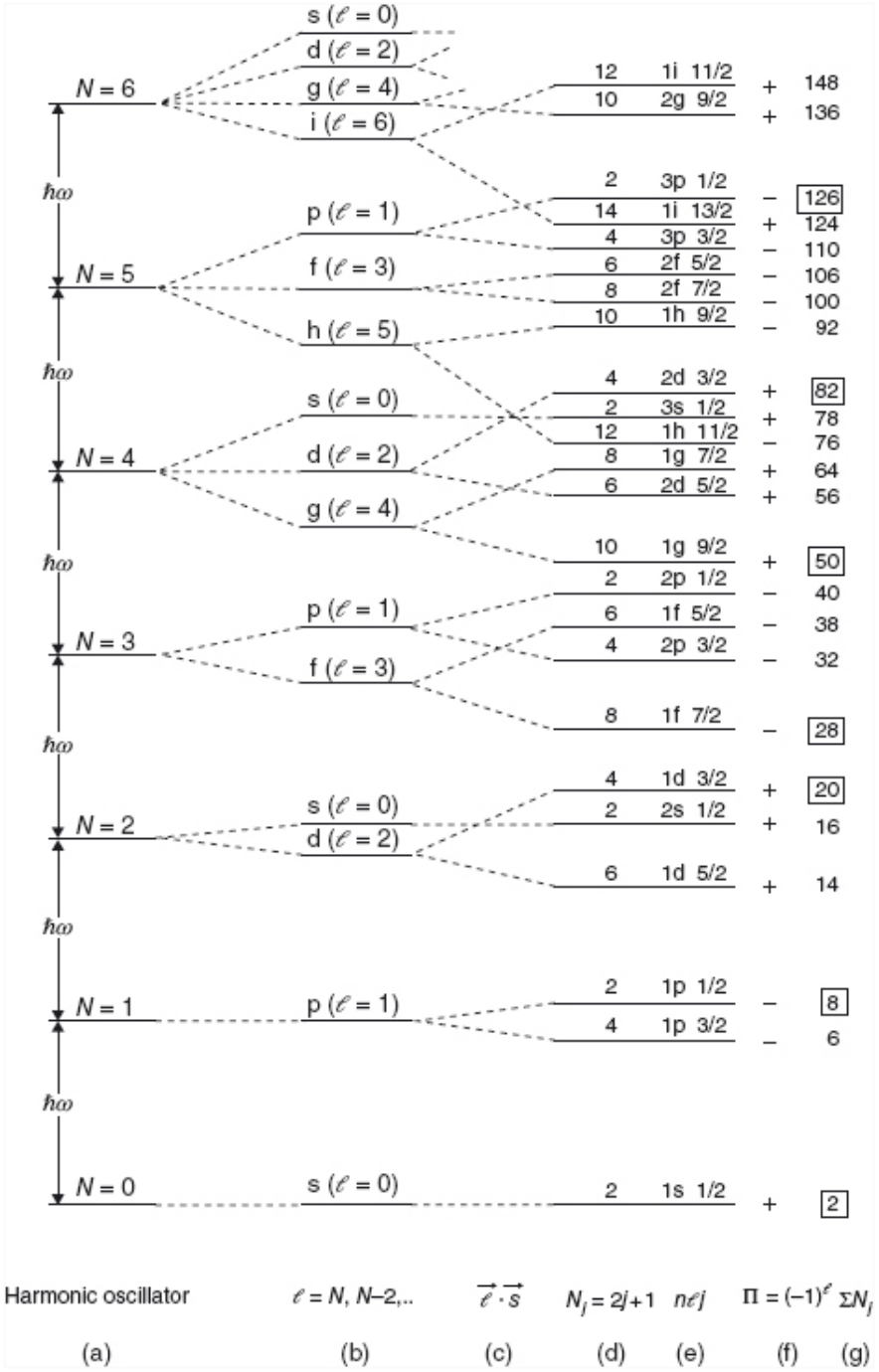
c) Spin-orbit term splits each state with $\ell > 0$ into two states given by $j = \ell - \frac{1}{2}$ or $j = \ell + \frac{1}{2}$

d) Number of identical particles that can occupy a state given by $2j + 1$

e) Single-particle spectroscopic notation $n\ell j$

f) Parities of single-particle levels given by $\pi = (-1)^{\ell}$

g) Total number of nucleons that can fill the states up to the given level

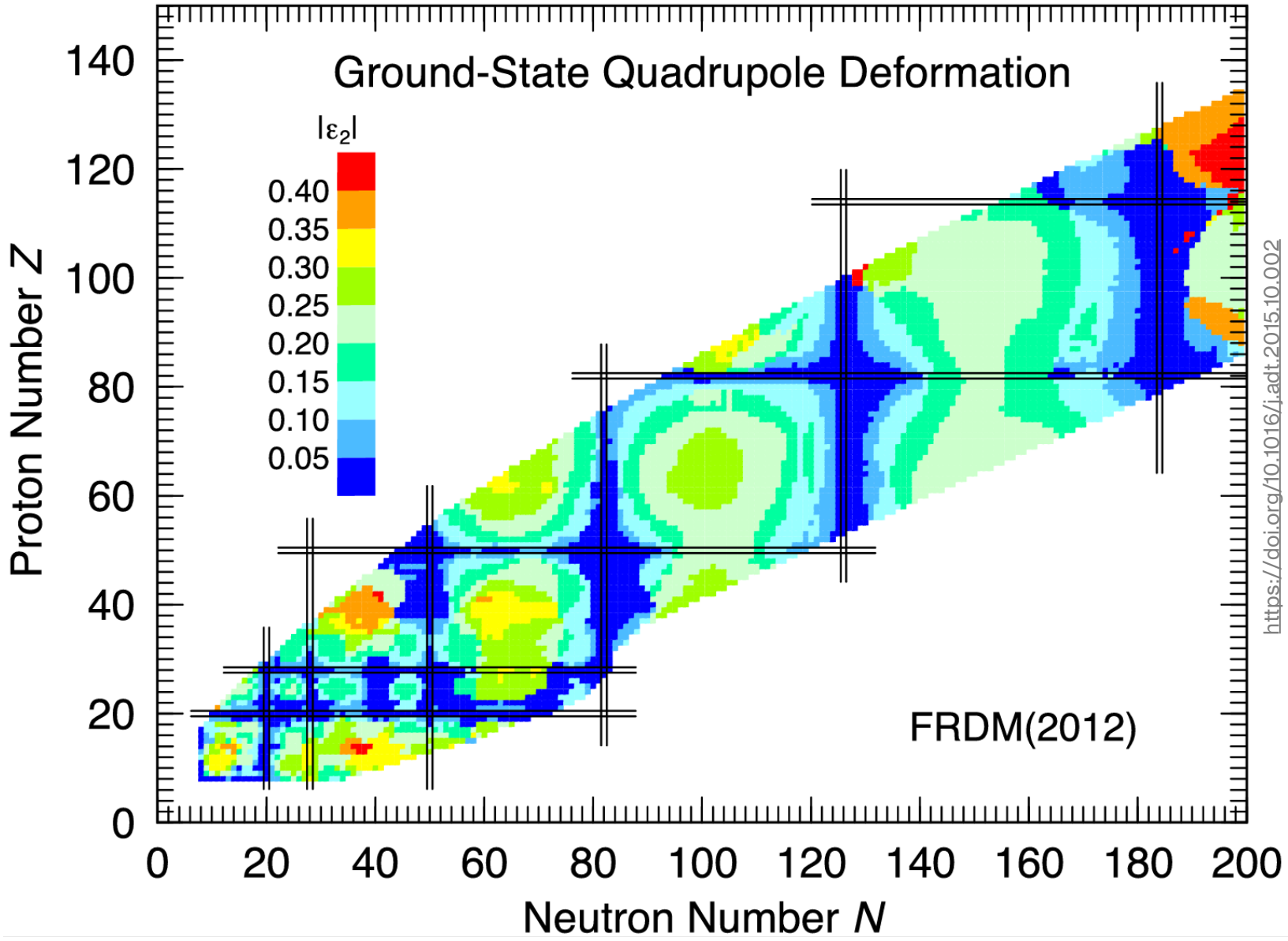


Nuclear Physics of Stars, C. Iliadis

Collective Model

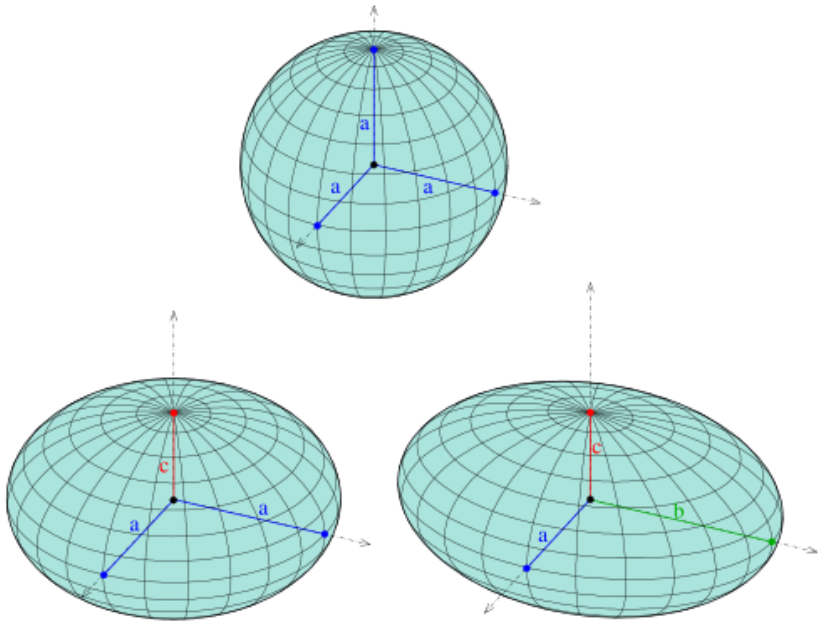
- Nucleons residing in well-defined orbitals in the nucleus can be understood with *relatively* simple math using the Shell Model
  - Properties of the nucleus are dominated by one or two unpaired nucleons
- However, the bulk of the nucleons, which may even be hundreds, only contribute to the central potential and not the nuclear properties
- In reality, we cannot ignore the core nucleons because they give rise to large-scale macroscopic behavior that can be different from the behavior of single particles
  - Independent vs. collective motion
- Two important collective motions:
  - Rotation of deformed nuclei
  - Vibrations of the nuclear shape

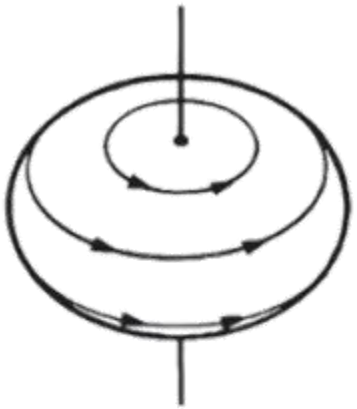
Predicted Regions of Deformation



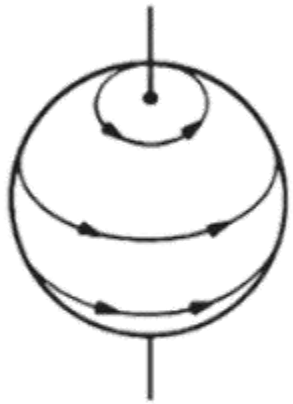
Nuclear Shapes Reminder

- Treat the nucleus as an ellipsoid (3D object formed by rotation of an ellipse around one of its two major axes)
- **Prolate deformation:** one axis is longer relative to the other two (rugby ball ”rug ball”)
- **Oblate deformation**: one axis is shorter than the other two (pancake)

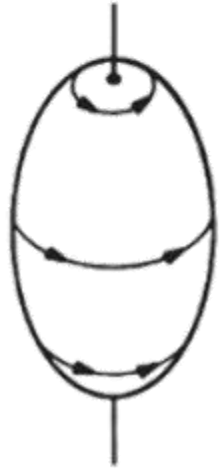




Oblate nucleus
Quadrupole moment < 0



Spherical nucleus
Quadrupole moment = 0

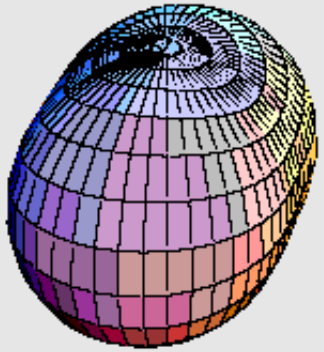


Prolate nucleus
Quadrupole moment > 0

Collective Motion

- Rotational and vibrational motion have unique signatures in excited states
- Relative energies of rotation vs. vibration can be inferred from their geometries
  - Rotational frequency should be similar to $\omega_r \propto \frac{1}{R^2}$ because the moment of inertia is $I = \frac{L}{\omega}$ and $I \propto MR^2$
  - Vibrational frequency should be $\omega_v \propto \frac{1}{(\Delta R)^2}$ because it's like an oscillator
  - $\omega_r \ll \omega_v$

Rotational Motion



https://web2.ph.utexas.edu/~coker2/index.files/rotation2.htm

Vibrational Motion



https://web2.ph.utexas.edu/~coker2/index.files/rotational.htm

Rotational Motion: Rigid Rotor

- Energy associated with a rotating object is: $\boldsymbol{E_{rot} = \frac{1}{2} I\omega^2}$
- But, we need this to reflect a quantum mechanical system where $\boldsymbol{J = I\omega}$
- So, $\boldsymbol{E_{rot} = \frac{1}{2}\frac{J^2}{I}}$ and J is quantized making the rotational energy:

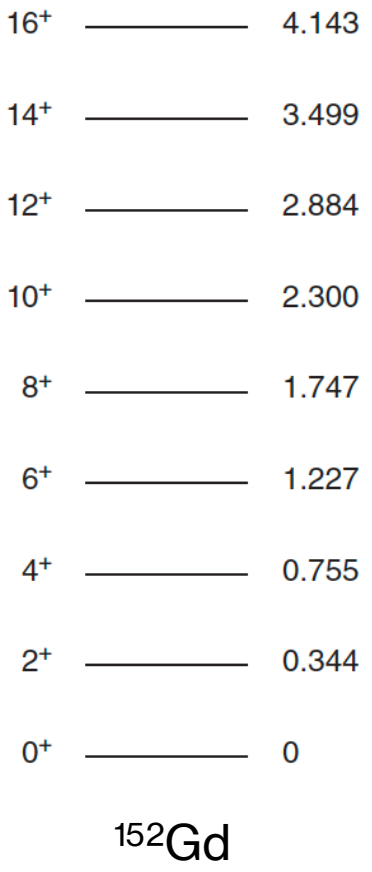
$$\boldsymbol{E_{rot} = \frac{j(j+1)\hbar^2}{2I}}$$

- The rotating nucleus will have excited states spaced by $\boldsymbol{j(j+1)}$ corresponding to rotation
- To calculate moment of inertia, you can consult formulae in the Bohr & Mottelson textbook
  - Generally, for a solid constant-density ellipsoid,

$$\boldsymbol{I_{rigid} = \frac{2}{5} MR^2\left(1 + 0.31\beta + 0.44\beta^2 + \ \ldots\right)}$$

Rotational Bands

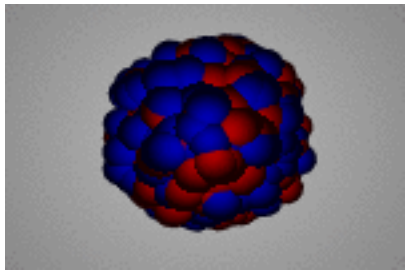
- Sequences of excited states
- $\boldsymbol{E_{rot} = \frac{j(j+1)\hbar^2}{2I}}$, so for a given I, $\Delta \boldsymbol{E_{rot} \propto j(j+1)}$
- Note that parity must be maintained because rotation is symmetric upon reflection. 0^+ ground states can only have $\boldsymbol{j = 0^+, 2^+, 4^+}$, ...
- Note that this interpretation works for even-even nuclei. Odd nuclei are another deal....
- Rotational bands would be challenging to observe without measuring a full level scheme with coincidences



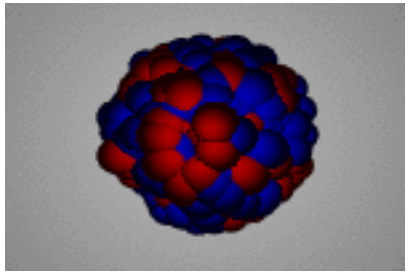
Vibrational Motion

- Considering the nucleus as a liquid drop, the nuclear volume should be able to vibrate
- Imagine the surface of the nucleus vibrating harmonically
  - We can parameterize the surface oscillations in terms of the spherical harmonic functions
  - We label these quanta (phonons) as λ
- Several multipoles are possible
  - Monopole: in & out motion ($\lambda = 0$)
    - "breathing mode"
  - Dipole: sloshing back & forth ($\lambda = 1$)
  - Quadrupole: alternating compression & stretching ($\lambda = 2$)
  - Octupole: alternating pinching on one end and then the other ($\lambda = 3$)
- Not all nucleons need to move together

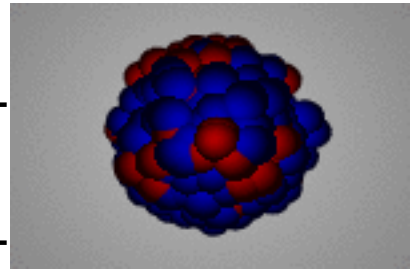
monopole



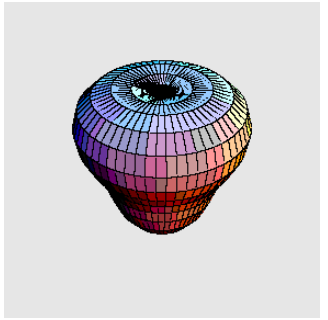
dipole



quadrupole



octupole



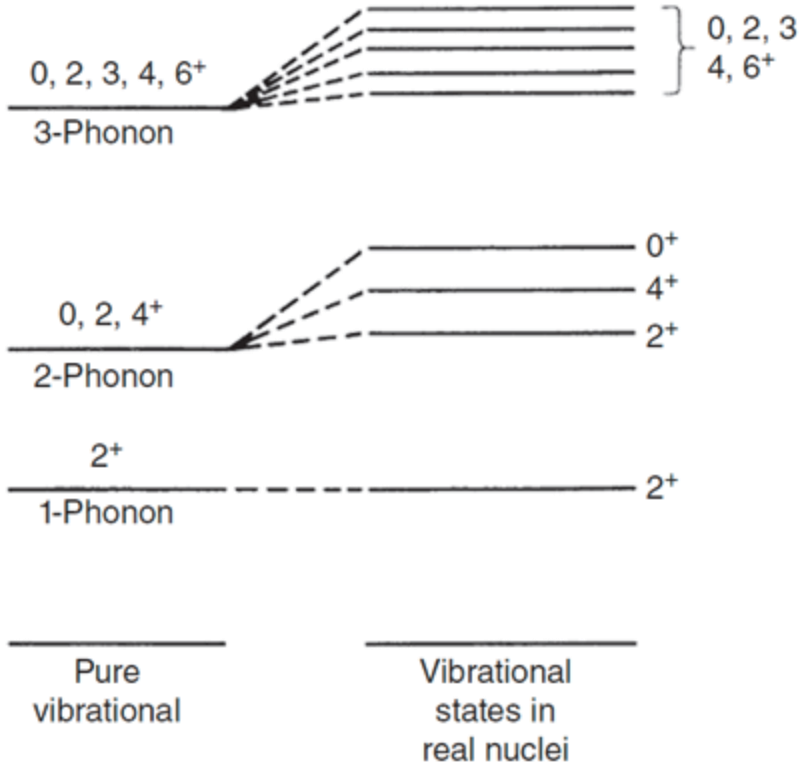
https://web-docs.gsi.de/%7Ewolle/TELEKOLLEG/KERN/index-s.html

- Recall that the energy levels of a quantum mechanical harmonic oscillator are all equally spaced having energies

  $E_N = (N + \frac{1}{2})\hbar\omega_0,\ N = 0, 1, 2, \ldots$

- There is some oscillating deviation from the default shape and a restoring force attempts to return the situation to the default shape

- Just as the quantum harmonic oscillator eigenvalues are quantized, so too will the energy levels for different quanta (phonons) of a vibrational mode



- Even-even nuclides have 0^+ ground states and thus for a $\lambda = 2$ vibration, $\mathrm{N} = 2$ excitations will maintain the symmetry of the wave function.

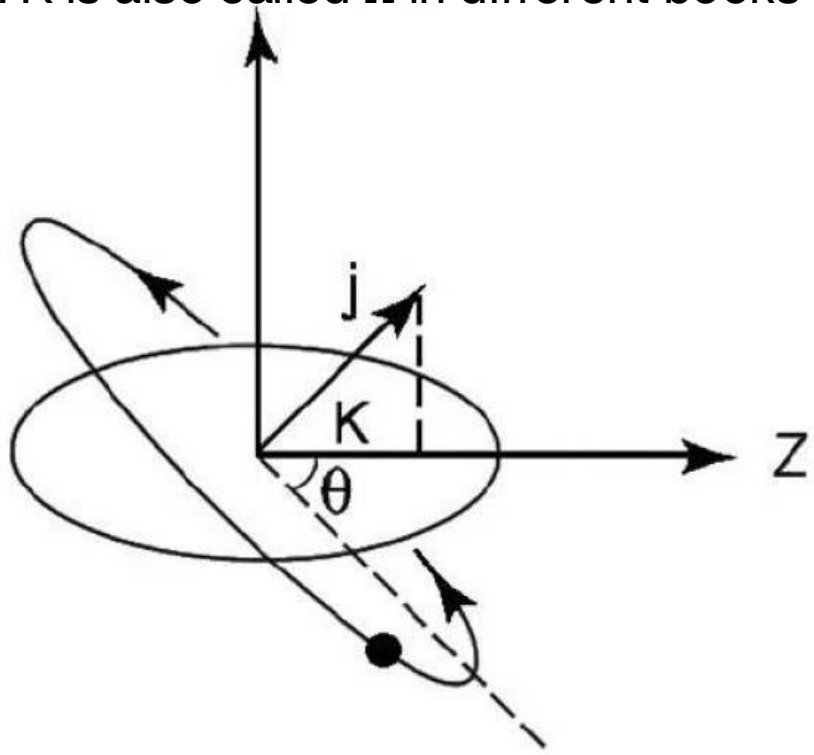
- Therefore, the first vibrational state will be a 2^+

- We can excite an independent quadrupole vibration by adding a second phonon

- The second phonon will build excitations on the first, coupling to either $0^+, 2^+$, or 4^+

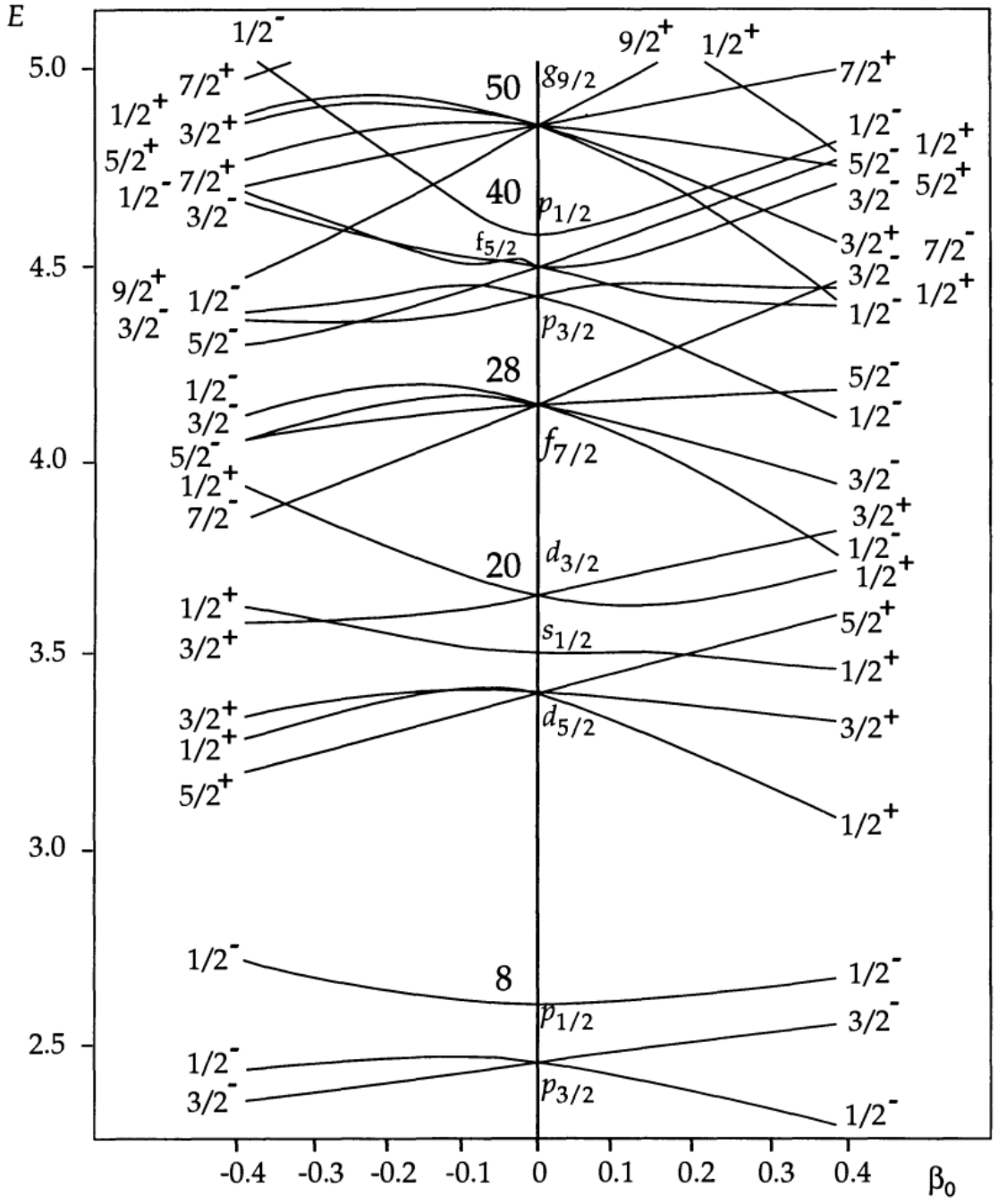
Nilsson Model

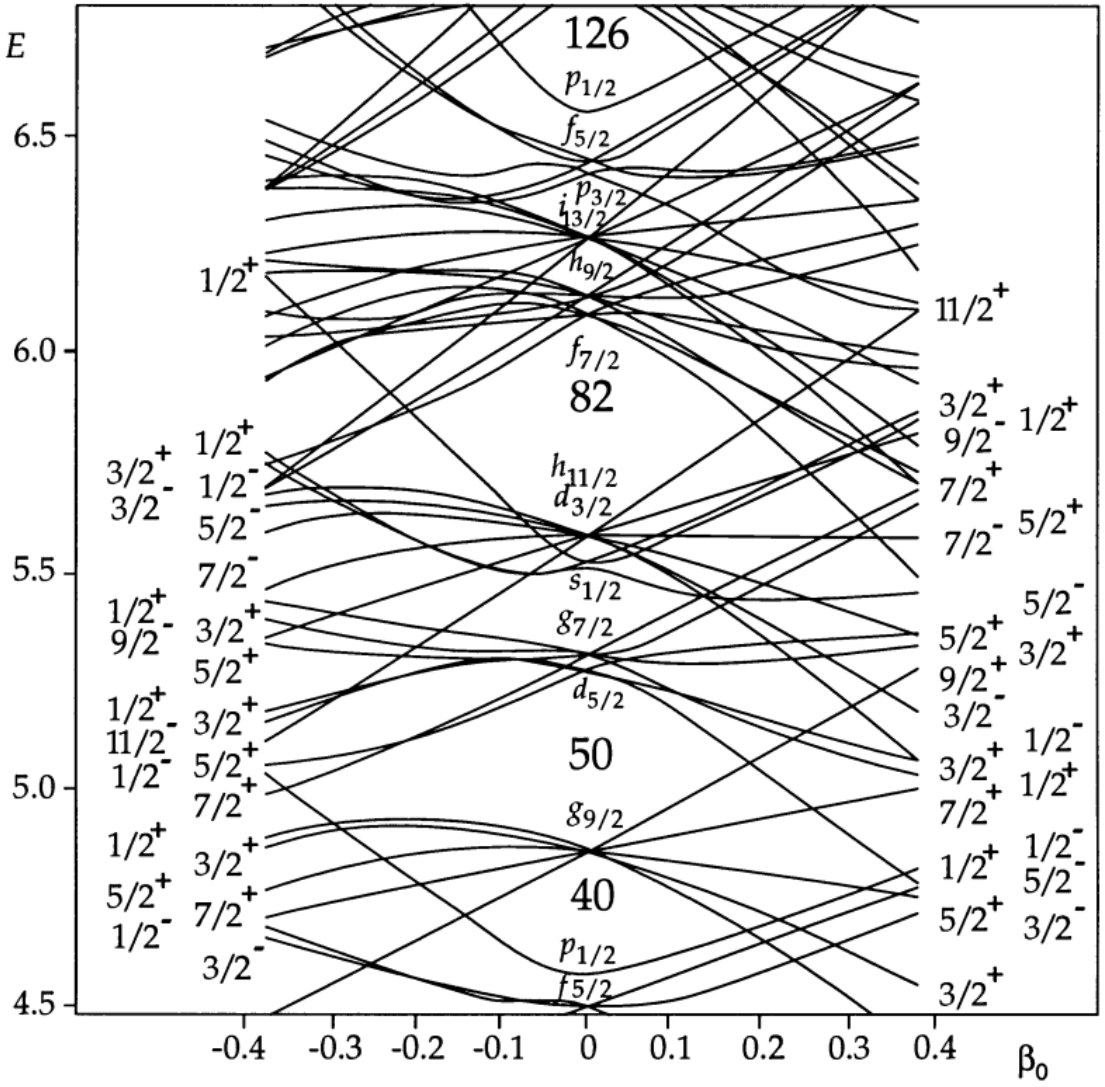
Note: K is also called Ω in different books



R.Casten, Nuclear Structure from a Simple Perspective (1990)

- Until now, we've discussed the two extremes of nuclear structure: **independent particle motion** and **collective motion**
- We can't use our traditional shell model (that we know and love) due to deformation
- Consider a deformed nucleus with axial symmetry that has a single unpaired nucleon orbiting the nucleus
  - We will only consider axially symmetric deformation in this class for simplicity
- The nucleon has some spin, j, which has a projection onto the symmetry axis, K
- j has $\frac{2j+1}{2}$ possible projections of K using a deformed Harmonic Oscillator potential
- Therefore, each single-particle state from our Shell Model splits into multiple states, identified by their K, each of which can contain two particles (spin up and down) for that projection

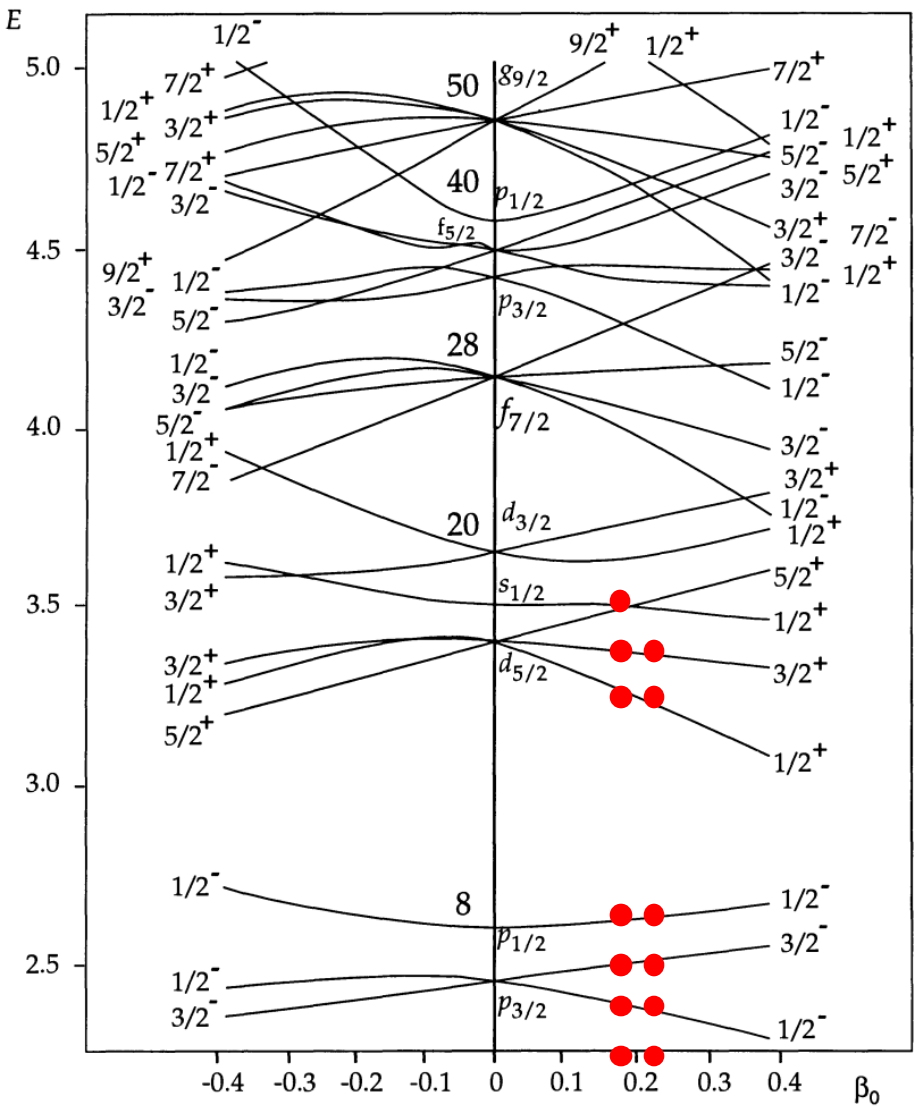




Nuclear Models by Greiner and Maruhn

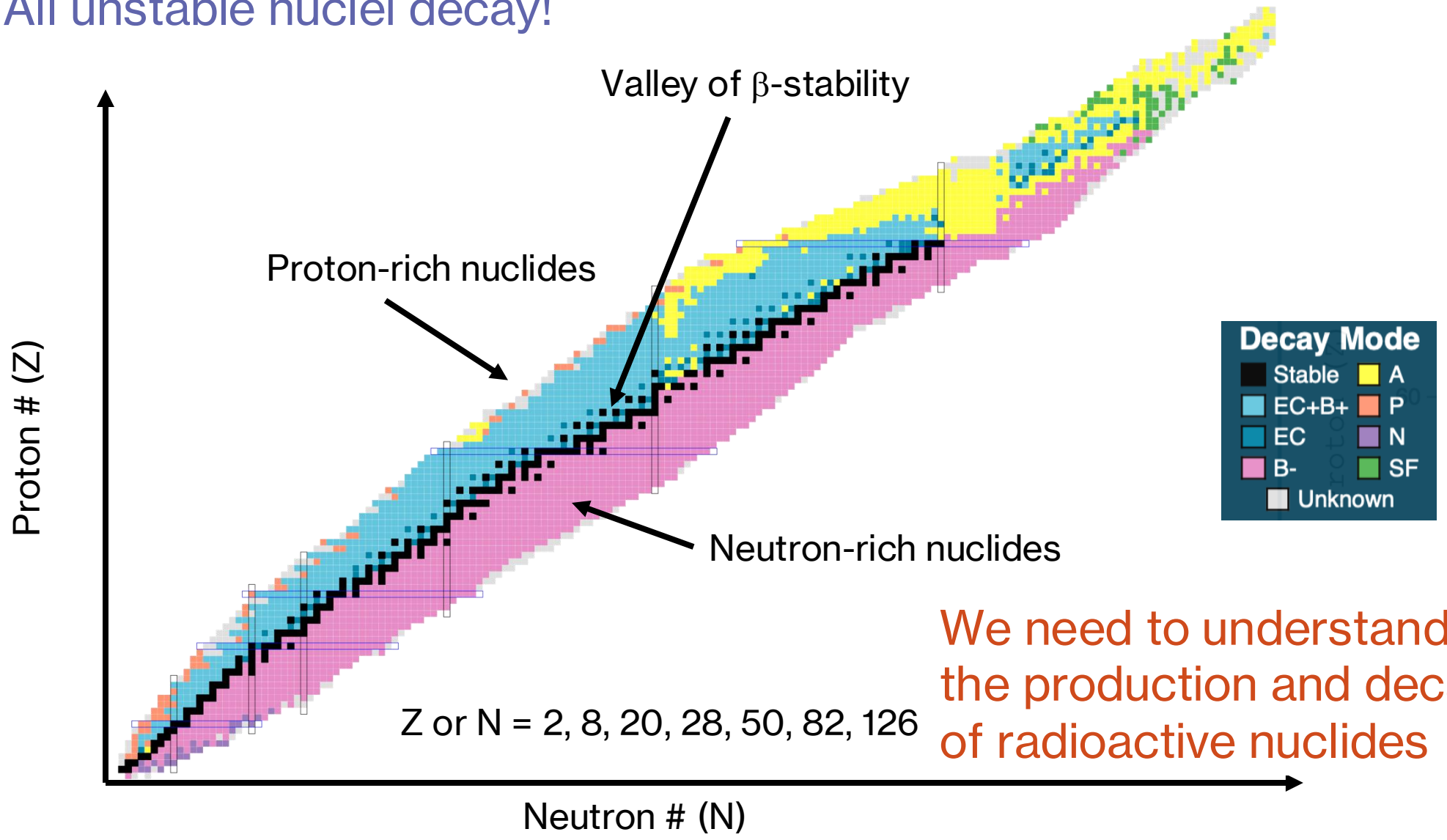
Nilsson Model Example

- Consider ^{25}Al, which we expect to have $\beta_2 \approx 0.2$
- There are 13 protons and 12 neutrons, so the unpaired proton will determine the J^π
- Filling the single particle levels as usual
- So, we predict $J^\pi = \frac{5}{2}^+$
- For the first excited state, it is probable that the proton will move to the $\frac{1}{2}^+$ level above
- This agrees with data



Radioactive Decay in the Chart of the Nuclides

All unstable nuclei decay!

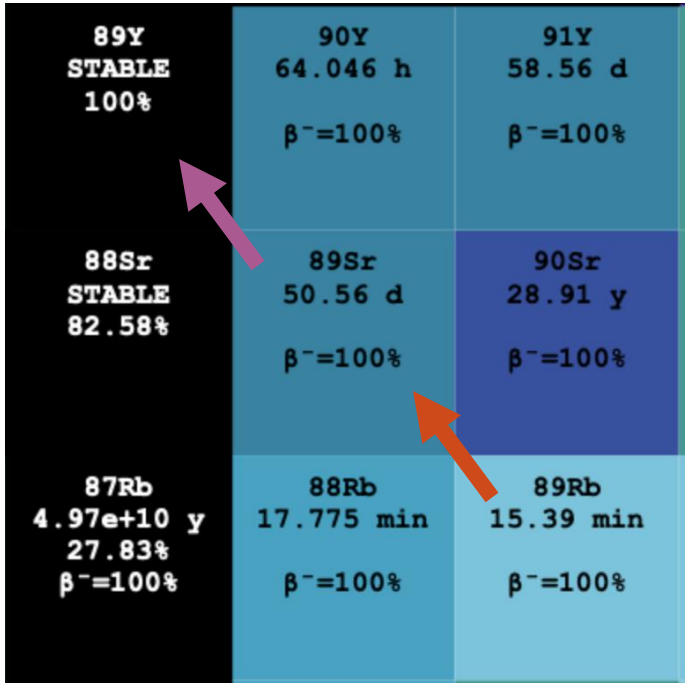


We need to understand the production and decay of radioactive nuclides

Radioactive Decay

- When an isotope decays, it does not disappear but is transformed into a new nuclear species of lower total energy and often differing in $\mathrm{Z}, \mathrm{A}, \mathrm{J}, \pi$, and so on.
- So far, we've only considered the case where the sample decays, but we haven't talked about the formation daughter, grand-daughter, and so on

$$\boldsymbol{Parent \rightarrow Daughter \rightarrow Granddaughter}$$
$$\boldsymbol{Parent \rightarrow Child \rightarrow Grandchild}$$



Characterized by decay half-life:

$$t_{1/2} = \frac{ln(2)}{\lambda}$$

- Consider the case when a radionuclide (1) decays with decay constant λ_1, forming a daughter nucleus (2), which in turn decays with decay constant λ_2
- Essentially, we have $1 \rightarrow 2$ (nuclide 1 decaying into nuclide 2 which then in turn also decays)
- We write the decay expression in terms of the production and depletion of (2)

$$\textbf{rate of change in } 2 = \textbf{production rate of } 2 - \textbf{decay rate of } 2$$

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$$

N_1 and N_2 are the amounts of (1) and (2) at time **t**

$$dN_2 + \lambda_2 N_2 dt = \lambda_1 N_1 dt$$

Recall: $N_1 = N_1^0 e^{-\lambda_1 t}$

$$dN_2 + \lambda_2 N_2 dt = \lambda_1 N_1^0 e^{-\lambda_1 t} dt$$

First-order differential equation

Can be solved as:

$$N_2(t) = \overbrace{\frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 \left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right)}^{\textbf{Grow-in of daughter}} + N_2^0 e^{-\lambda_2 t}$$

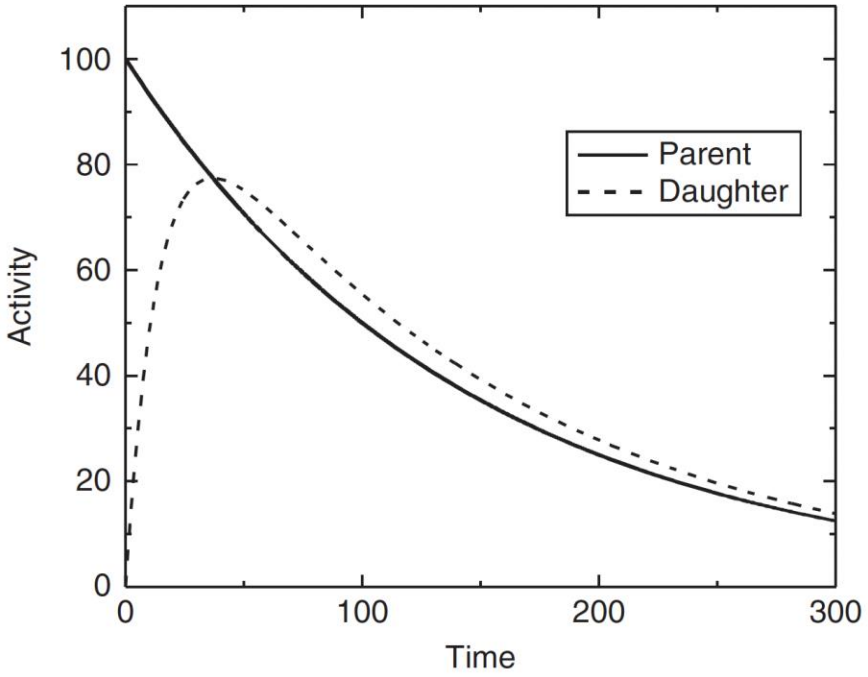
Decay of daughter

- We can write the activity as

$$A_2(t) = \frac{\lambda_1\lambda_2}{\lambda_2 - \lambda_1} \mathrm{N}_1^0\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right) + A_2^0 e^{-\lambda_2 t}$$

- The total activity is the combination of the parent and daughter where the exact time dependence relies on the relationship between λ_1 and λ_2
- How do we determine the maximum daughter activity?
- Use the condition that $\frac{dN_2}{dt} = 0$ at the maximum.
- The derivate of the activity equation solved for time yields:

$$t_{max} = \frac{ln(\lambda_2/\lambda_1)}{\lambda_2 - \lambda_1}$$



Initial activity of the daughter is zero, rising to a maximum, and eventually decreasing

- For the case where the daughter is stable, $\lambda_2 = 0$ so our expression reduces to

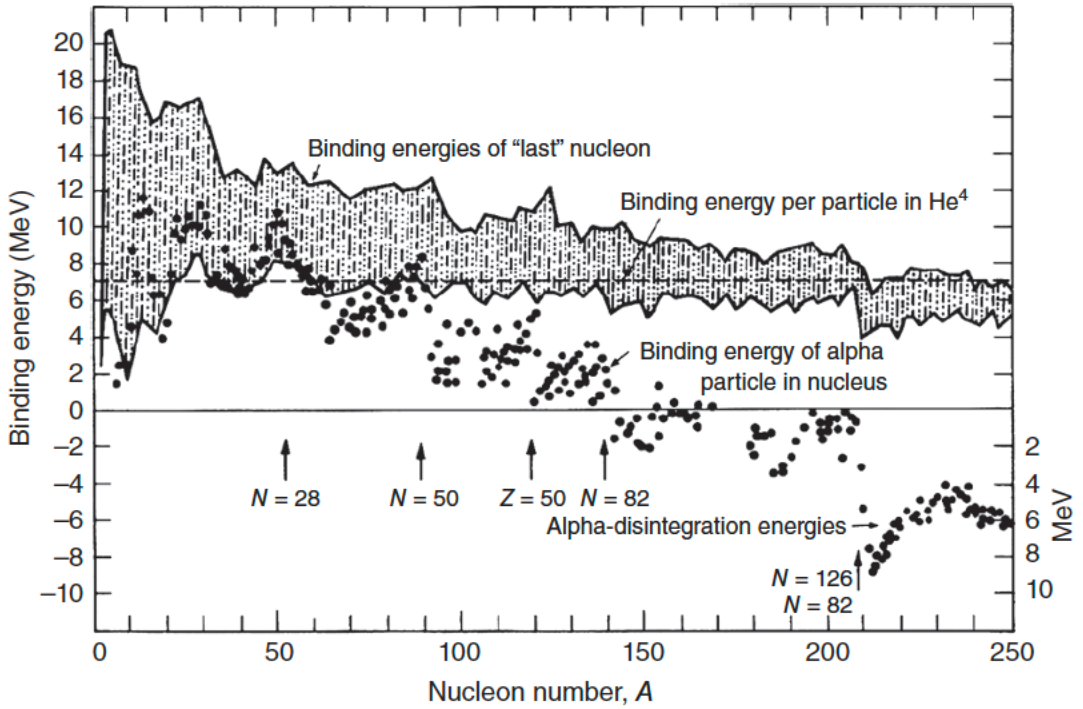
$$N_2(t) = N_1^0\left(1 - e^{-\lambda_1 t}\right)$$

You can also imagine expanding this to $1 \rightarrow 2 \rightarrow 3$ ...
These expressions are often referred to as the **Bateman Equations**

α-Decay Basics

- Alpha decay is a spontaneous process by which heavy nuclei emit an $\boldsymbol{\alpha}$ particle moving the system to a lower energy state
- The process of alpha-decay can be written as: $\mathbf{{}^{A}_{Z}Z_{N} \rightarrow {}^{A-4}_{Z-2}Z_{N-2} + {}^{4}_{2}He_{2} + Q_{\alpha}}$
  - Example: ${}^{238}\mathrm{U} \rightarrow {}^{234}\mathrm{Th} + {}^{4}\mathrm{He} + Q_{\alpha}$
- The value of $\boldsymbol{Q_{\alpha}}$ (Q-value) is positive, or endothermic, for spontaneous $\boldsymbol{\alpha}$ decay

$$\boldsymbol{Q_{\alpha} = \Delta_{parent} - \Delta_{daughter} - \Delta_{^4He}}$$



-Why does it make sense for $\boldsymbol{Q_{\alpha}}$ to be positive?

-In terms of the SEMF, losing 2 protons lowers the Coulomb energy, doesn't impact asymmetry or pairing, and minimally changes the surface and volume terms (per A)

-While the last nucleon is tightly bound, so is a nucleon a ^{4}He nucleus

-It is not nearly as bound as the nucleus itself, so $\boldsymbol{\alpha}$ emission is energetically favorable above A~150

Energy Emitted from α Decay

- When an α is emitted, it will share energy with the heavy recoil – what is the relationship between kinetic energy and Q_α?
- Conservation of energy and momentum:

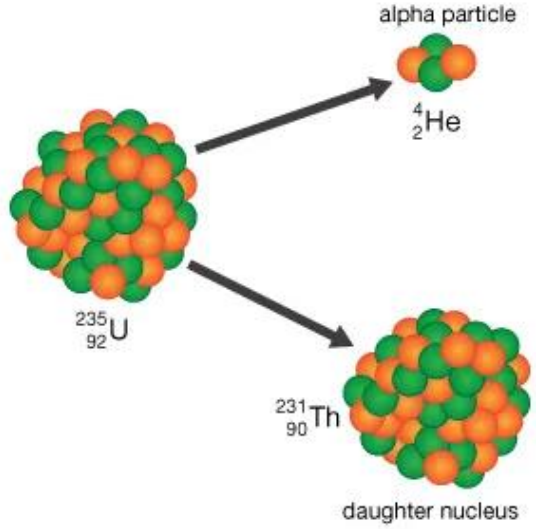
$$\vec{\mathbf{p}}_{\mathbf{parent}} = \vec{\mathbf{p}}_{\mathbf{daughter}} + \vec{\mathbf{p}}_{\boldsymbol{\alpha}}$$

- But $\vec{\mathrm{p}}_{\mathrm{parent}} = 0$ because the parent is at rest. So, the daughter and α will move in opposite directions: $\vec{\mathrm{p}}_{\mathrm{daughter}} = -\vec{\mathrm{p}}_{\alpha}$
- Then for energy:

$$\frac{\boldsymbol{p_{parent}^2}}{\boldsymbol{2A_{parent}}} + \boldsymbol{Q_\alpha} = \frac{\boldsymbol{p_{daughter}^2}}{\boldsymbol{2A_{daughter}}} + \frac{\boldsymbol{p_\alpha^2}}{\boldsymbol{2A_\alpha}} = \frac{\boldsymbol{p_\alpha^2}}{\boldsymbol{2A_{daughter}}} + \frac{\boldsymbol{p_\alpha^2}}{\boldsymbol{2A_\alpha}} = \frac{\boldsymbol{A_\alpha}}{\boldsymbol{A_{daughter}}}\boldsymbol{KE_\alpha} + \boldsymbol{KE_\alpha}$$

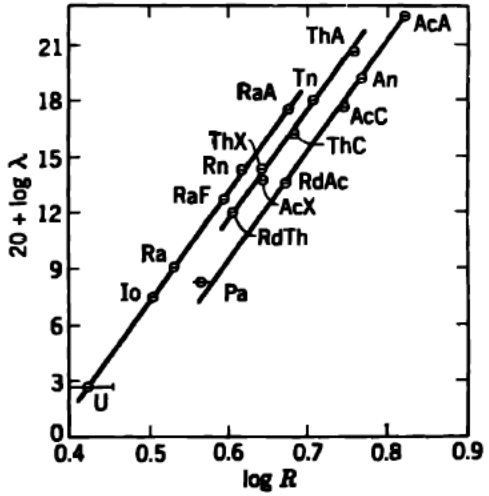
- $Q_\alpha = \frac{A_\alpha + A_{daughter}}{A_{daughter}} KE_\alpha = \frac{A_{\mathrm{parent}}}{A_{daughter}} KE_\alpha$
- So, $\boldsymbol{KE_\alpha = \frac{A_{daughter}}{A_{parent}} Q_\alpha}$

This is a small effect overall



Geiger-Nuttall Relation

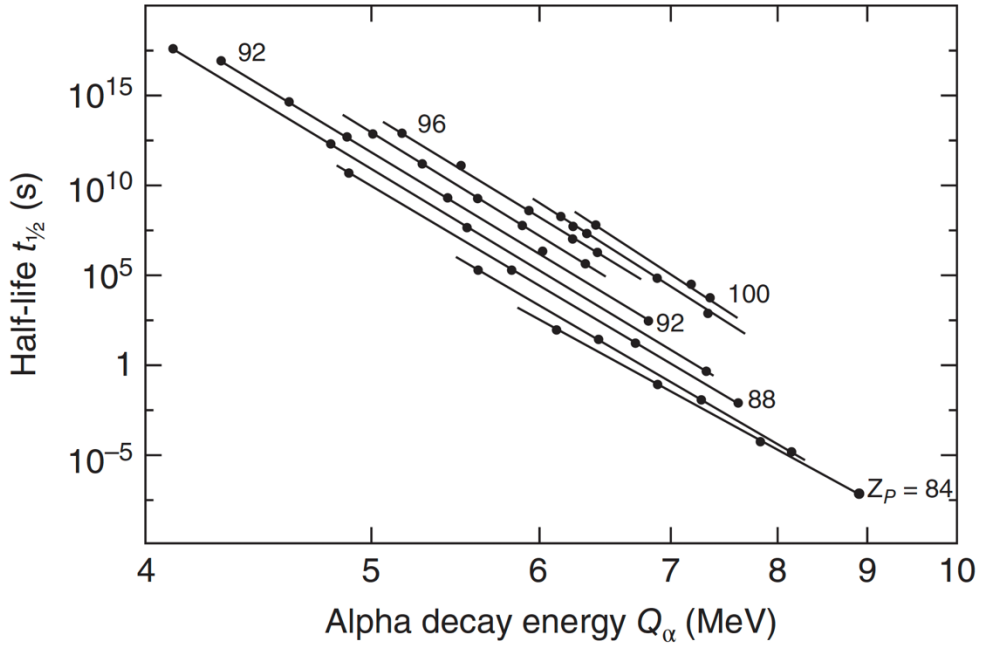
Geiger & Nuttall, Philisoph. Mag. (1912)



- In an early effort to characterize α decay, Geiger & Nuttall compared the range of α particles in a material vs. $t_{1/2}$ of the α emitter (source)
  - They discovered a linear relationship in log-log space
- Using Q_α instead of range, we get the Geiger-Nuttall relation:

$$log_{10}(t_{1/2}) = A + \frac{B}{\sqrt{Q_\alpha}}$$

A and B have Z dependence



- α energy strongly impacts the $t_{1/2}$
- For a $\times 2$ increase in Q_α nearly 20 orders of magnitude decrease in $t_{1/2}$

Tunneling through a thick barrier

- The transmission coefficient is $\mathrm{T} = e^{-2\mathrm{G}}$, where

This is pretty complicated

$$2G = \frac{2}{\hbar}\sqrt{2\mu Q_\alpha}\int_b^R \left(\frac{b}{r}\right)^{1/2} dr = 2\frac{e^2}{\hbar c}Z_\alpha Z_{daughter}\sqrt{\frac{2\mu c^2}{Q_\alpha}}\left[\cos^{-1}\left(\sqrt{\frac{R}{b}}\right) - \sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}\right]$$

- Conveniently, most of the time $b \gg R$ so $\left[cos^{-1}\left(\sqrt{\frac{R}{b}}\right) - \sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}\right]$ reduces to $\frac{\pi}{2}\left[1-\frac{4}{\pi}\sqrt{\frac{R}{b}}\right]$
- So, the Gamow factor becomes

$$2G \approx 2\pi\frac{e^2}{\hbar c}Z_\alpha Z_{daughter}\sqrt{\frac{\mu c^2}{2Q_\alpha}}\left(1-\frac{4}{\pi}\sqrt{\frac{r_0\left(A_\alpha^{1/3}+A_{daughter}^{1/3}\right)}{\frac{Z_\alpha Z_{daughter}}{Q_\alpha}e^2}}\right)$$

- Since half-life is inversely proportional to the tunneling probability, $t_{1/2} \propto e^{2G} \propto e^{z/\sqrt{Q_\alpha}}$
- This is exactly what the Geiger-Nuttall relation already told us!

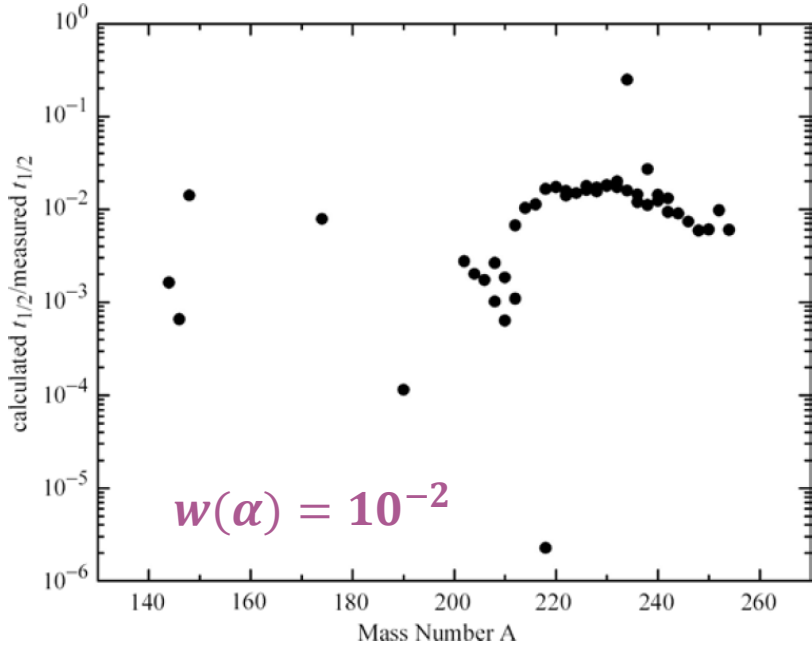
Estimating the α-decay rate (λ)

- The decay constant is a product of the frequency that an α will bombard the barrier f, the tunneling probability T, and the probability of forming an α within the nucleus $w(\alpha)$

$$\lambda = fTw(\alpha)$$

- We already know that

$$T = exp\left(-2\pi\frac{e^2}{\hbar c}Z_\alpha Z_{daughter}\sqrt{\frac{\mu c^2}{2Q_\alpha}}\left(1-\frac{4}{\pi}\sqrt{\frac{r_0(A_\alpha^{1/3}+A_{daughter}^{1/3})}{\frac{Z_\alpha Z_{daughter}}{Q_\alpha}e^2}}\right)\right)$$



- f is one over the time it takes for the α to travel across the nucleus

$$f = \frac{v_\alpha}{2R} = \sqrt{\frac{2(V_0+Q_\alpha)}{\mu}}\frac{1}{2r_0(A_\alpha^{1/3}+A_{daughter}^{1/3})}$$

What is β decay? There are 3 types

$\bar{\nu}_e$ - anti-neutrino
ν_e - neutrino

Nucleus perspective

$\beta^-: {}^{A}_{Z}X_N \longrightarrow {}^{A}_{Z+1}Y_{N-1} + e^- + \bar{\nu}_e + Q_{\beta-}$

$\beta^+: {}^{A}_{Z}X_N \longrightarrow {}^{A}_{Z-1}Y_{N+1} + e^+ + \nu_e + Q_{\beta+}$

EC: ${}^{A}_{Z}X_N + e^- \longrightarrow {}^{A}_{Z-1}Y_{N+1} + \nu_e + Q_{EC}$

Nucleon perspective

$\beta^-: n \longrightarrow p + e^- + \bar{\nu}_e + Q_{\beta-}$

$\beta^+: p \longrightarrow n + e^+ + \nu_e + Q_{\beta+}$

EC: $p + e^- \longrightarrow n + \nu_e + Q_{EC}$

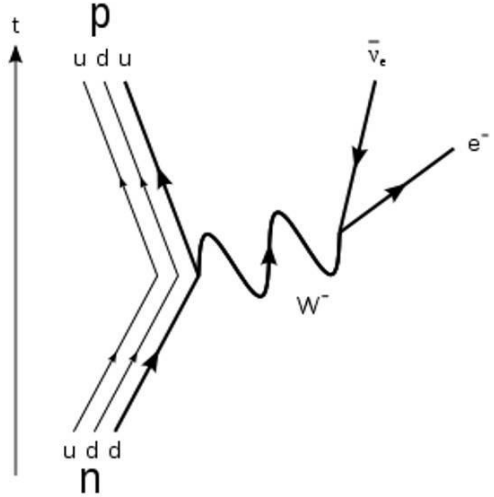
Quark perspective

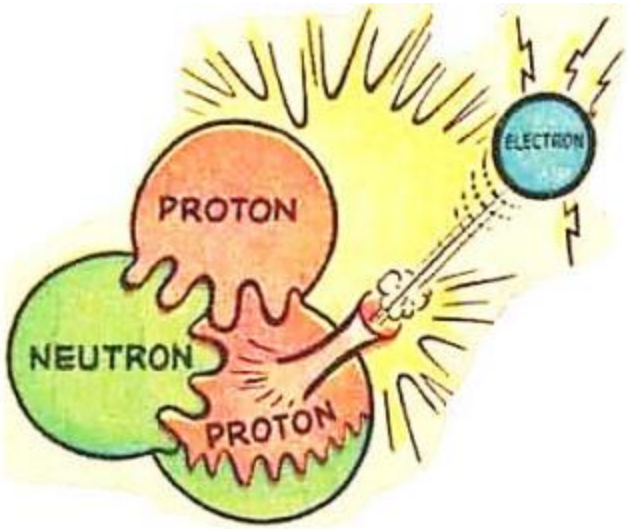
$\beta^-: d \longrightarrow u + e^- + \bar{\nu}_e + Q_{\beta-}$

$\beta^+: u \longrightarrow d + e^+ + \nu_e + Q_{\beta+}$

EC: $u + e^- \longrightarrow d + \nu_e + Q_{EC}$

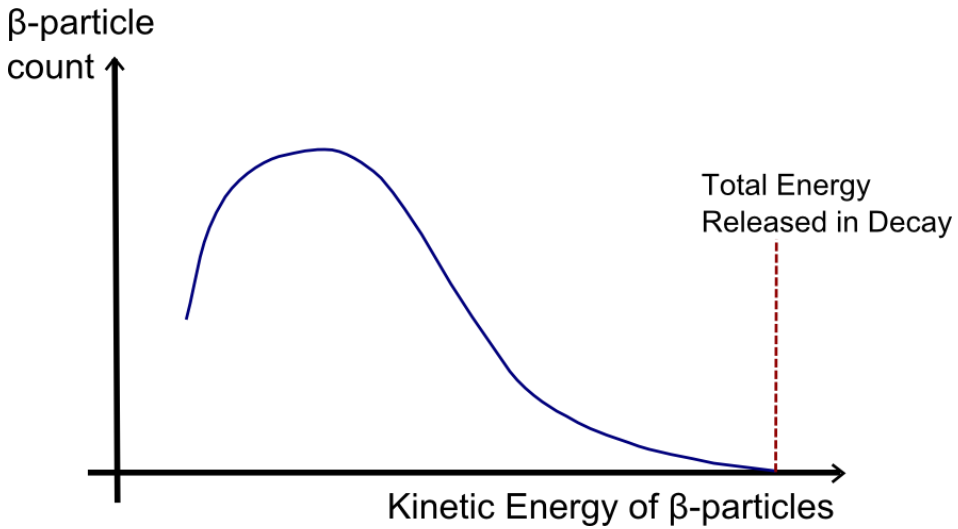
- More complicated than α decay
  - Three-body final state
- β decays are mediated by the *weak interaction*
  - W^+, W^-, Z^0 bosons are the mediators
  - Responsible for transmutation of neutron into a proton plus an electron and an electron anti-neutrino...





β-Decay Energy Spectrum

- Early experiments investigating the "β ray" showed that it was not emitted with a singular energy, like the "α ray", but rather in a continuum of energies
  - The maximum energy is the decay Q value
- Furthermore, the reaction $n \rightarrow p + e^-$ doesn't conserve spin!
  - $J_n = J_p = J_e = \frac{1}{2}$..... So, $0 \leq J_P + J_e \leq 1 \neq \frac{1}{2}$



- Pauli postulated the existence of a third hypothetical particle (the neutrino) to explain the shape of the β-decay spectrum.
  - He proposed this in a letter to Lise Meitner
- Given the considerations, it was postulated that υ was a spin-1/2 particle (fermion) that is massless (true enough for our purposes) and electrically neutral
- In one of his last works before switching to experimental work, Fermi postulated that nucleons could act as sources and sinks of neutrinos, which was the only successful theory of interactions between quantum particles (in 1934)

We know this is true today, but *Nature* actually rejected Fermi's paper for being "too remote from physical reality"

Total β-decay rate and *ft* values

- Integrating the form of the b-decay spectrum over all electron momenta yields the decay constant:

$$\lambda = \frac{g^2 |M_{fi}|^2}{2\pi^3 \hbar^7 c^3} \int_0^{p_{max}} F(Z_d, p_e)\, p_e^2 (Q - T_e)^2 dp_e$$

Assuming a constant NME

- So this is hard… but we can define the Fermi Integral and use numerical methods to integrate
- Dimensionless Fermi Integral:

$$f(Z_d, Q) \propto \int_0^{p_{max}} F(Z_d, p_e)\, p_e^2 (Q - T_e)^2 dp_e$$

- Then, our expression for the decay rate becomes

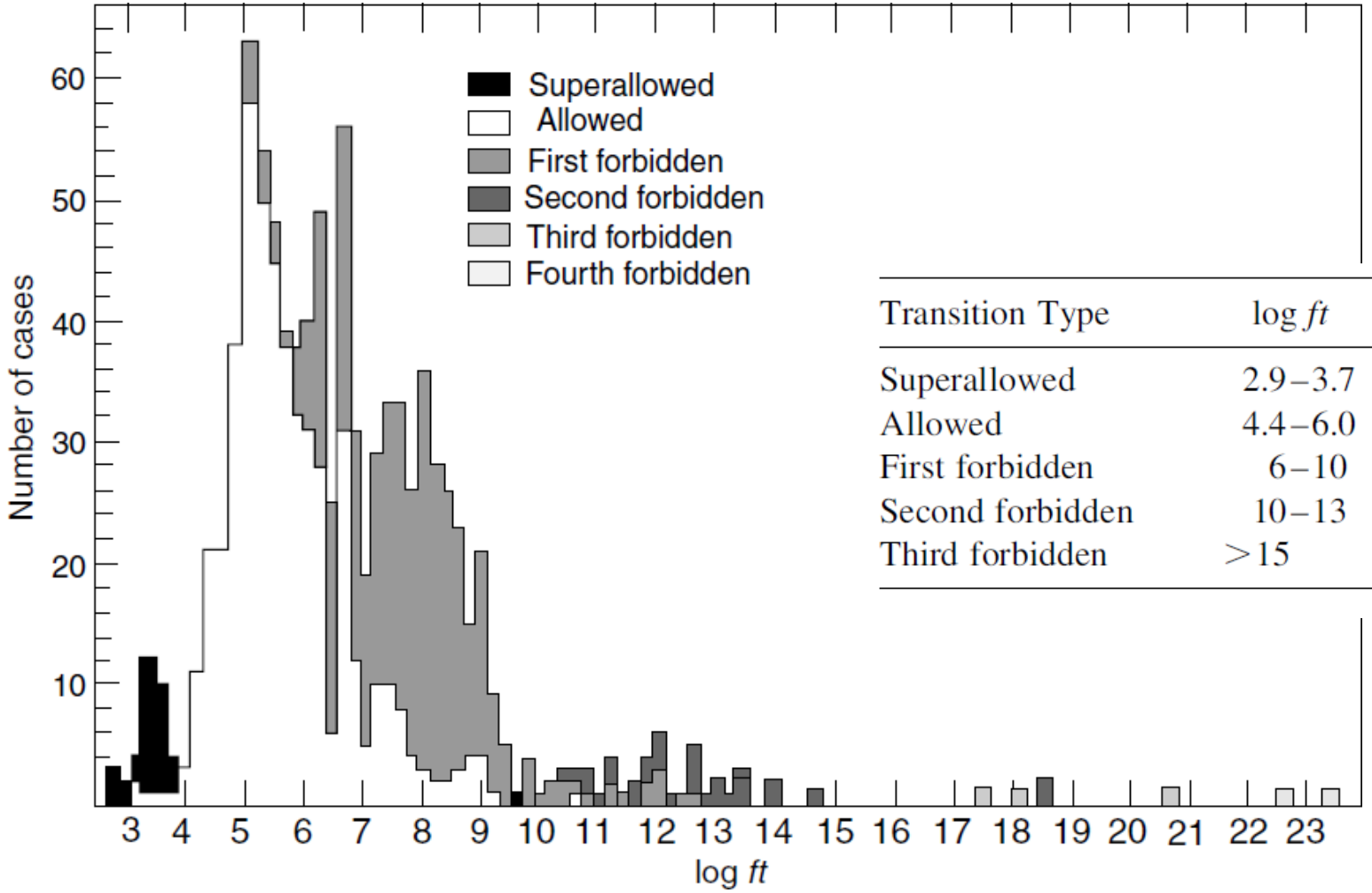
$$\lambda = \frac{g^2 |M_{fi}|^2}{2\pi^3 \hbar^7 c^3} f(Z_d, Q) = \frac{ln(2)}{t_{1/2}}$$

Decay Selection Rules

- Ignoring higher order terms in the Taylor expansion omits the possibility of angular momentum transfer.
- If the angular momentum transfer does occur, higher-order terms need to be included, and the matrix element can no longer be independent of p_e
- For cases where $\Delta \mathrm{J} > 1$ and/or $\Delta\pi = \mathrm{yes}$, the leading order overlap $\left|\boldsymbol{M_{fi}}\right|^{\mathbf{2}} = 0$ so a higher-order term is needed
- This combined with if there is or is not a change in parity is referred to as how *forbidden* a decay is... though it's not really forbidden
  - **Super−Allowed :** $\mathbf{0^+ \rightarrow 0^+}$
  - **Allowed:** $\mathbf{0^+ \rightarrow 1^+}$ **or** $\boldsymbol{\Delta J = 0\ or\ 1}$ **and** $\boldsymbol{\Delta\pi = no}$
  - **First Forbidden:** $\boldsymbol{\Delta J = 0\ or\ 1}$ **and** $\boldsymbol{\Delta\pi = yes}$
  - **Second Forbidden:** $\boldsymbol{\Delta J = 2}$**and** $\boldsymbol{\Delta\pi = no}$

Empirical log(ft) Values and Selection Rules

W. Meyerhof, Elements of Nuclear Physics (1967)



Loveland, Morrissey, & Seaborg, Modern Nuclear Chemistry (2006)

Transition Type	log *ft*	L_β	$\Delta\pi$	Fermi ΔI	Gamow–Teller ΔI
Superallowed	2.9–3.7	0	No	0	0
Allowed	4.4–6.0	0	No	0	0, 1
First forbidden	6–10	1	Yes	0, 1	0, 1, 2
Second forbidden	10–13	2	No	1, 2	1, 2, 3
Third forbidden	>15	3	Yes	2, 3	2, 3, 4

Electron Capture

- When the decay energy is < 1.022 MeV ($2m_e c^2$) but larger than 0, the β decay of a proton-rich nucleus to its daughter must be via electron capture (EC)
- For decay energies > 1.022 MeV, $\beta^+$and EC compete.
- In EC decay, only one particle is emitted from the nucleus (the neutrino).
- Assuming a zero neutrino rest mass, the decay constant for EC can be written as

$$\lambda_{\mathrm{EC}} = \frac{\boldsymbol{g^2 \left| M_{fi} \right|^2} \mathrm{T}_{\boldsymbol{v}}^2}{\boldsymbol{2\pi^3 \hbar^7 c^3}} |\phi_{\mathrm{K}}(0)|^2$$

- The K electron wave function can be written as $\phi_{\mathrm{K}}(0) = \frac{1}{\sqrt{\pi}} \left(\frac{\mathrm{Z m_e e^2}}{4\pi\epsilon_0 \hbar^2} \right)^{3/2}$
- So, the ratio between $\beta^+$and EC goes as $\frac{\lambda_{\mathrm{EC}}}{\lambda_{\beta+}} \propto \mathrm{Z}^3$ meaning that EC is more probable for heavy nuclei

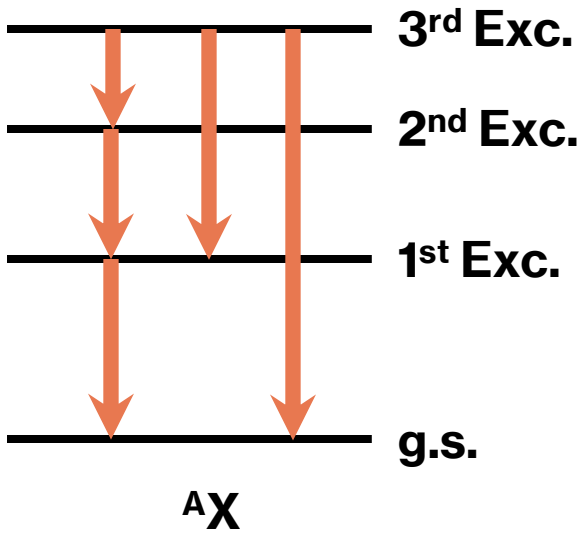
What is Gamma Decay?

- γ decay occurs when a nucleus is in an excited state and releases its excess energy via emission of electromagnetic radiation, a photon or γ ray

$$ {}^{A}_{Z}X^{*}_{N} \longrightarrow {}^{A}_{Z}X_{N} + \gamma $$

- No change in Z, A and * denotes an excited state of the nucleus
- γ-ray energies can range from few keV to 10s of MeV
- γ-decay lifetimes are extremely short ($\tau \lesssim$ femtoseconds)

  except for isomeric states
- Can proceed from to the ground state or any number of high-lying states to low-lying states



Classifications of Gamma Decay

- Therefore, the angular momentum carried away by the photon will be

$$\left|(\boldsymbol{I_i}-\boldsymbol{I_f})\right| \leq \ell \leq \left|(\boldsymbol{I_i}+\boldsymbol{I_f})\right|$$

- The *multipolarity* of the photon is a label for the amount of angular momentum carried by the photon
- The nomenclature is that a photon with ℓ units of angular momentum is called a 2^{ℓ} −pole photon
  - $\ell = 1$ is called a dipole photon, $\ell = 2$ is called a quadrupole photon, …
- Parity depends on ℓ and the decay type
- Electric and Magnetic multipoles of the same order always have opposite parity!

Radiation Type	Name	$\ell = \Delta I$	$\Delta\pi$
E1	Electric dipole	1	(Yes)
M1	Magnetic dipole	1	(No)
E2	Electric quadrupole	2	(No)
M2	Magnetic quadrupole	2	(Yes)
E3	Electric octupole	3	(Yes)
M3	Magnetic octupole	3	(No)
E4	Electric hexadecapole	4	(No)
M4	Magnetic hexadecapole	4	(Yes)

No $0 \rightarrow 0$ transitions!

	maximum spin change	nuclear parity change
Eℓ:	ℓ	if ℓ is odd
Mℓ:	ℓ	if ℓ is even

Electromagnetic Transition Rates

- Without proof and extensive calculus, we can obtain an expression for the electromagnetic decay rate:

$$\boldsymbol{\lambda(\ell, I_i, \pi \rightarrow I_f, \pi)} = \frac{\boldsymbol{8\pi(\ell+1)}}{\boldsymbol{\ell[(2\ell+1)!!]^2}} \frac{\boldsymbol{k^{2\ell+1}}}{\hbar} \boldsymbol{B(\ell, I_i, \pi \rightarrow I_f, \pi)}$$

$\boldsymbol{k}$ is the photon wave number, !! is a double factorial ($\boldsymbol{\ell = 2, (2\ell+1) = 5, 5!! = 5 \times 3 \times 1 = 15}$), and $\boldsymbol{B(\ell, I_i, \pi \rightarrow I_f, \pi)}$ is the reduced transition probability, a matrix element for the reduced nuclear wave functions.

- The reduced transition probabilities get really complicated really quickly and are challenging to evaluate.
- Victor Weisskopf derived a general expression for the reduced transition probability with the assumption that the transition results from a change of a single particle inside a nucleus with a uniform density of radius $R = r_0 A^{1/3}$

Weisskopf Estimates

- His expression for ***electric*** multipole radiation is called the *Weisskopf Single Particle Limit*

$$B_{sp}(E,\ell) = \frac{1}{4\pi}\left[\frac{3}{(\ell+3)}\right]^2 (r_0)^{2\ell} A^{2\ell/3} e^2 fm^{2\ell}$$

- Similarly, the ***magnetic*** multipole radiation can be expressed

$$B_{sp}(M,\ell) = \frac{10}{\pi}\left[\frac{3}{(\ell+3)}\right]^2 (r_0)^{2\ell-2} A^{(2\ell-2)/3} e^2 fm^{(2\ell-2)}$$

- These can be substituted into the expression for transition rate to obtain numerical estimates
- Weisskopf estimates are generally good within a few orders of magnitude of the real answer so γ decay constants are often quoted as a ratio to this estimate in *Weisskopf Units [w.u.]*

Weisskopf Estimates

- Combining the expressions for transition rate and reduced transition probability for an E1 transition, we obtain

$$\boldsymbol{\lambda_{sp}(\ell, I_i, \pi \rightarrow I_f, \pi)} = \frac{\boldsymbol{8\pi(\ell+1)}}{\boldsymbol{\ell[(2\ell+1)!!]^2}} \frac{\boldsymbol{k^{2\ell+1}}}{\boldsymbol{\hbar}} \frac{\boldsymbol{1}}{\boldsymbol{4\pi}} \left[\frac{\boldsymbol{3}}{\boldsymbol{(\ell+3)}}\right]^{\boldsymbol{2}} \boldsymbol{(r_0)^{2\ell} A^{2\ell/3} e^2 f m^{2\ell}}$$

- Substituting in $\ell = \mathbf{1}$

$$\boldsymbol{\lambda_{sp}(E1)} = \frac{\boldsymbol{8\pi(2)}}{\boldsymbol{[3!!]^2}} \frac{\boldsymbol{k^3}}{\boldsymbol{\hbar}} \frac{\boldsymbol{1}}{\boldsymbol{4\pi}} \left[\frac{\boldsymbol{3}}{\boldsymbol{4}}\right]^{\boldsymbol{2}} \boldsymbol{(r_0)^2 A^{2/3} e^2 f m^2}$$

which yields

$$\boldsymbol{\lambda_{sp(E1)} = 1.03 \times 10^{14}\ E_\gamma^3 A^{2/3} s^{-1}}$$

- Repeat this for all multipoles

Internal Conversion

- A process that competes with γ-ray decay in which an excited nucleus interacts electromagnetically with an orbital electron and ejects it.

$$E_{IC} = E_{transition} - E_{electron\ binding\ energy}$$

- Competition between γ-ray decay and IC is given by an internal conversion coefficient

$$\alpha = \frac{\lambda_{IC}}{\lambda_{\gamma}} \qquad \frac{\#\ \text{of internal conversion decays}}{\#\ \text{of}\ \gamma - \text{ray decays}}$$

- The total decay rate is: $\lambda = \lambda_{IC} + \lambda_{\gamma} = \lambda_{\gamma}(1 + \alpha)$

- α depends on the density of electrons near the nucleus, and IC coefficients have been calculated in Atomic Physics

$$\alpha(E, \ell) = \frac{Z^3}{n^3}\left(\frac{\ell}{\ell + 1}\right)\left(\frac{e^2}{4\pi\epsilon_0\hbar c}\right)^4\left(\frac{2m_e c^2}{E}\right)^{\ell+5/2}$$

$$\alpha(M, \ell) = \frac{Z^3}{n^3}\left(\frac{e^2}{4\pi\epsilon_0\hbar c}\right)^4\left(\frac{2m_e c^2}{E}\right)^{\ell+3/2}$$

- n is the principal quantum number of the electron being ejected
- Atomic orbitals K, L, M, N, … correspond to n = 1, 2, 3, 4, …
- $\frac{e^2}{4\pi\epsilon_0\hbar c}$ is the fine structure constant

Wrapping up!

- HW #3 posted and extended to Oct. 14th **at 5PM**
- Midterm is on **Oct. 7**
  - Bring one-page cheat sheet, calculator
- Office Hours Edwards 204, 10AM Friday
- **Next class**: Midterm Exam
- Questions?